Jordan University of Science and Technology
College of Computer Sciences & Information Technology

Smart Park

A project submitted
in partial fulfillment of the requirements for the degree of
Bachelor' in Software Engineering

by

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Committee Member’s First Name Middle Initial Last Name
Committee Member’s First Name Middle Initial Last Name

Jan/2025

Undertaking

This is to declare that the project entitled "Smart Park" is an original work done by undersigned, in partial fulfillment of the requirements for the degree "Bachelor in Software Engineering" at Software Engineering Department, College of Computer and Information Technology, Jordan University of Science and Technology.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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ABSTRACT

Many car owners find it hard to find available parking spaces in public garages due to the increasing number of cars and the lack of organized parking management. This project proposes a mobile application designed to simplify and modernize the parking process. Through the app, drivers can easily check real-time parking availability in nearby garages and reserve a spot in advance based on their preferred time. Garage owners can also manage their parking capacity and update availability efficiently. The system aims to reduce traffic congestion, minimize time wasted searching for parking, and provide a structured and reliable solution for both drivers and garage owners. This application ultimately offers a more organized, accessible, and user-friendly approach to parking management.

Acknowledgment

We would like to express our sincere gratitude and appreciation to "**Dr. Zakaria AL-Shara"** for his continuous guidance, valuable feedback, and unwavering support throughout the development of this project. His expertise, encouragement, and constructive suggestions played a crucial role in shaping our understanding and improving the quality of our work.

We are also thankful for his patience and dedication, which greatly contributed to overcoming challenges during the analysis, design, and implementation phases of the project. His academic insight and professional guidance have been a constant source of motivation.

Finally, we would like to extend our appreciation to our families and friends for their moral support and encouragement throughout this journey.

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CHAPTER 1: Project Overview, Vision, and Planning

1.1 Problem Statement

In many urban areas across Jordan, especially in major cities such as Amman and Irbid, traffic congestion has become a persistent challenge for both residents and visitors. One of the primary contributors to this congestion is the significant increase in the number of vehicles compared to the limited availability of organized parking spaces. Drivers often spend a considerable amount of time searching for empty spots in public garages or on crowded streets, which leads to wasted time, increased fuel consumption, and additional traffic pressure on already busy roads.

The lack of an efficient and structured parking management system further intensifies the problem. Most public garages do not provide real-time information about parking availability, forcing drivers to physically visit each location to check for free spaces. This unorganized approach results in frustration, random circulation of vehicles inside cities, and reduced overall transportation efficiency. Additionally, garage owners face difficulties in managing their spaces, tracking occupancy, and communicating availability to drivers in a clear and timely manner.

This project aims to address these issues by developing a mobile application that connects car owners with garage owners through a unified digital platform. The application allows car owners to view real-time information about available parking spaces in selected areas and reserve a spot at their preferred time. At the same time, garage owners can efficiently manage their parking capacity, update availability, and monitor reservations digitally. By transforming the parking process into a structured and technology-driven system, the project seeks to reduce traffic congestion, optimize parking resource usage, and enhance the transportation experience for citizens in Jordan.

1.2 Related Products:

Feature / System	ParkMobile	Spot Hero	Park Me	ParkOpedi	EastPrck	Smart Park
Real-time parking availabilityy	Yes	No	No	No	No	Full real-time updates by garage owners
Owner control over pricing	No	No	No	No	No	Yes
Parking reservationn	No	Yes	No	No	Yes, on supported cities	Yes
Main operating regions	USA	USA & CANDA	Global	Global	Europe	Jordan
Supports small local garages	No	No	No	No	No	Yes
Garage owner dashboard	No	No	No	No	No	Yes
Admin control panel	No	No	No	No	No	Yes
Arabic language support	No	No	No	No	No	Yes

Table 1. Related Products.

1.3 Product Vision

- **Proposed Solution:**

  - The proposed solution is a mobile-based parking management system designed to simplify and organize the process of finding and reserving parking spaces in Jordanian cities. The system allows car owners to view real-time parking availability in nearby garages, reserve a parking spot in advance, and navigate to the selected location. At the same time, garage owners receive a dedicated dashboard where they can manage their parking spaces, update availability, adjust pricing, and review reservation requests. The project also includes an admin dashboard to supervise system activity, approve garages, manage users, and ensure smooth operation.

- **Target Users / Stakeholders:** Who will use or benefit from the system?

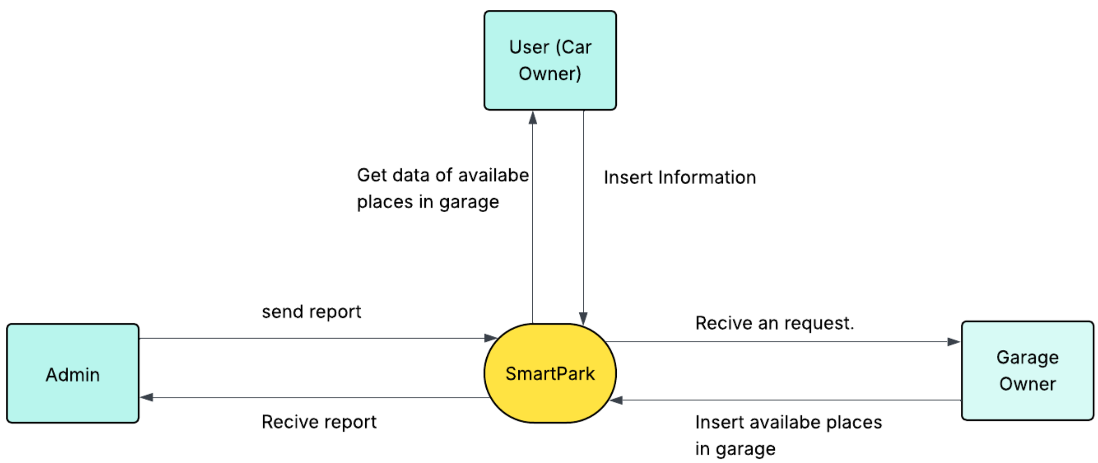


Figure 1. context diagram

Stakeholders are:
consist of three categories: Users, Garage Owner and Admin the system is used differently by each type of user, so their requirements are different:

1 - Car Owners (Users): People who need to park their cars in a garage. They use the app to see which garages have available spaces and to reserve a spot at the time they want.

2 - Garage Owners: People who manage or run a parking garage. They use the system to update the number of available spaces, manage reservations, and organize their garage in an easier way.

3- System Administrators: People who control and monitor the whole application. They use the admin dashboard to manage users, approve garages, and make sure everything in the system works correctly.

Product Vision Statement:
"Our system helps car owners find and reserve available parking spaces by providing a real-time, user-friendly platform that connects drivers with local garages, ensuring convenient parking at the right place and time."

Value Proposition:
Scope:
Included:

- User mobile application
- Garage owner management dashboard
- Admin web dashboard
- Real-time manual availability updates
- Parking reservation system

Not Included:

- IoT sensors or automated gates
- Online payment gateways (optional for future work)
- AI prediction features (planned for future versions)

Related products:

Existing apps like ParkMobile, SpotHero, ParkMe, and EasyPark operate mainly in developed countries and rely on advanced parking infrastructure. Your system is different because it is tailored for Jordan, supports Arabic, includes garage owner tools, operates without sensors, and offers an admin dashboard—making it more practical and suitable for local needs. 1.4 Project Objectives and Milestones

1.4 Project Objectives and Milestones

- The main objectives of this project are:

1. **Develop a mobile application** that allows car owners to check out available parking spaces in real time.
2. **Create a garage owner dashboard** to let garage managers update availability, manage reservations, and organize parking easily.
3. **Build an admin web dashboard** to control users, approve garages, and monitor the whole system.
4. **Reduce time wasted searching for parking** by providing a quick and organized reservation system.
5. **Improve parking management in Jordanian cities** by offering a digital, easy-to-use solution.
6. **Ensure a smooth user experience** through an intuitive interface for all three roles: users, garage owners, and administrators.

PROJECT SCHEDULE

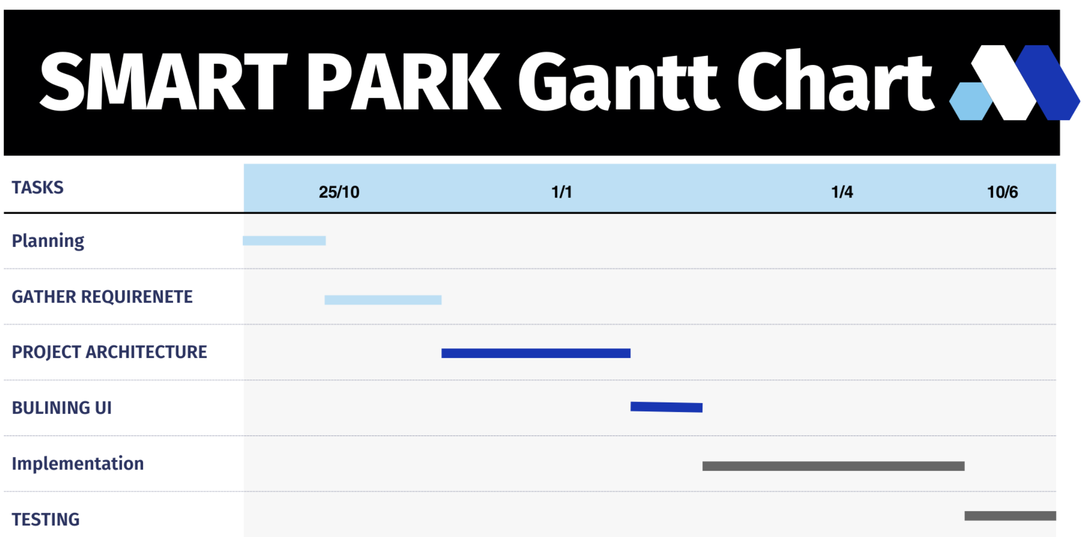


Figure 2. Gantt chart

1.5 Risk Assessment and Mitigation

Risk ID	Description	Impact	Mitigation Strategy
R1	Difficulty managing time and completing tasks within the planned schedule (Time Risk).	High	Create a weekly work plan, use a task-tracking tool, and review progress with the supervisor regularly.
R2	Changes in project requirements that may affect plans, timeline, or costs.	Medium	Document requirements clearly, get early approval from the supervisor, and limit major changes after development begins.
R3	Technical issues such as server crashes, network failures, or system errors.	High	Use reliable hosting, perform regular backups, conduct early testing, and apply error-handling mechanisms.
R4	Difficulty achieving the defined project goals, which may lead to incomplete or weak system performance.	Medium	Set realistic objectives, follow the design phase clearly, and perform continuous reviews and testing.
R5	Loss of project files or data due to device malfunction or viruses.	High	Use cloud storage (Google Drive / GitHub), maintain regular backups, and use updated antivirus software.
R6	Failure to meet user needs, resulting in low acceptance or dissatisfaction.	Medium	Conduct user feedback sessions, keep the UI simple, and ensure the system meets the core requirements.
R7	Discovering defects and unexpected bugs after delivery.	Medium	Perform thorough testing, create a bug list, fix issues before final submission, and run user acceptance tests.
R8	Integration issues between the mobile app and backend API.	High	Test each API endpoint early using Postman, integrate features step by step, and log all errors.

Table 2. Risk Assessment and Mitigation

CHAPTER 2: Product Features and Requirements

2.1 Functional Features

In gathering requirement process we have used these techniques

1- Interviews: we conducted interviews with individuals who are targeted by the application, whether through friends or colleagues in the university or by contacting groups or organizations working in the field of parking.

2- Brainstorming: we used brainstorming techniques with my team based on some information to gather ideas and suggestions about the application and its requirements, as well as the targeted audience.

3- Survey: we created a survey that ask users some questions to know the need of people about car parking, I got some good ideas from Survey responses, and this is a survey response:
Survey Link:

Response:

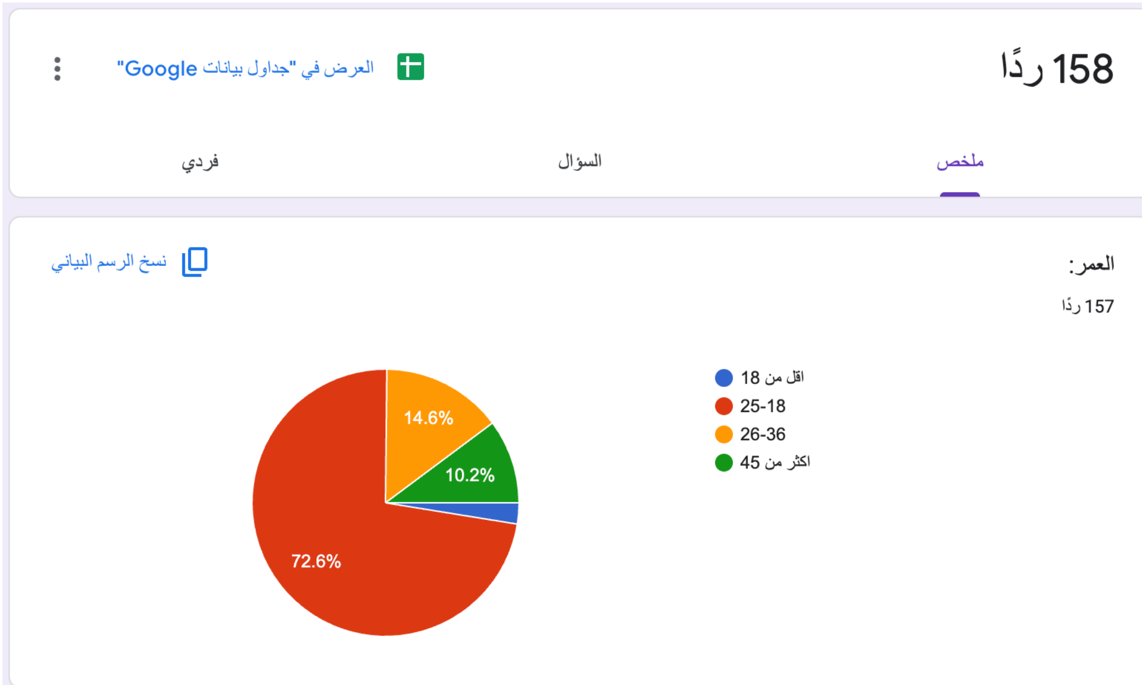
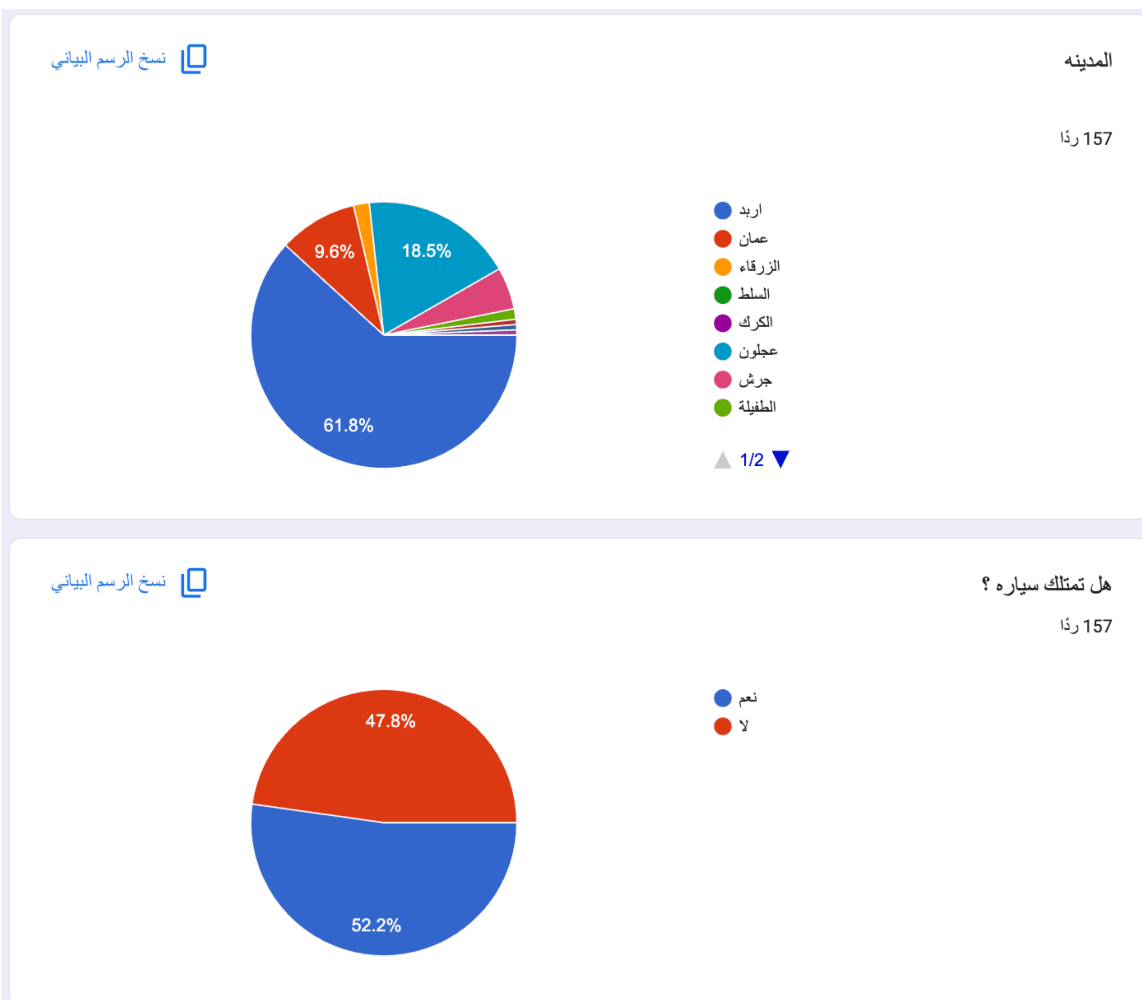
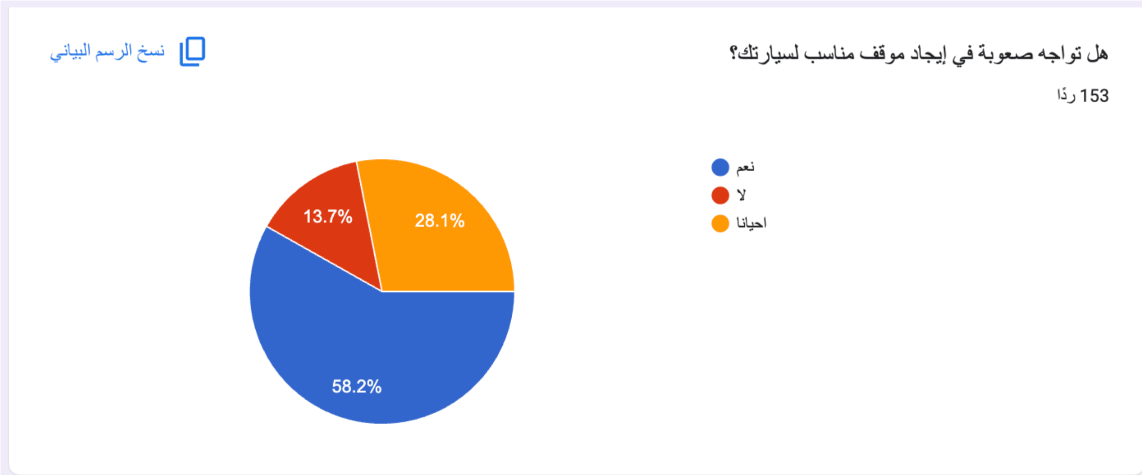
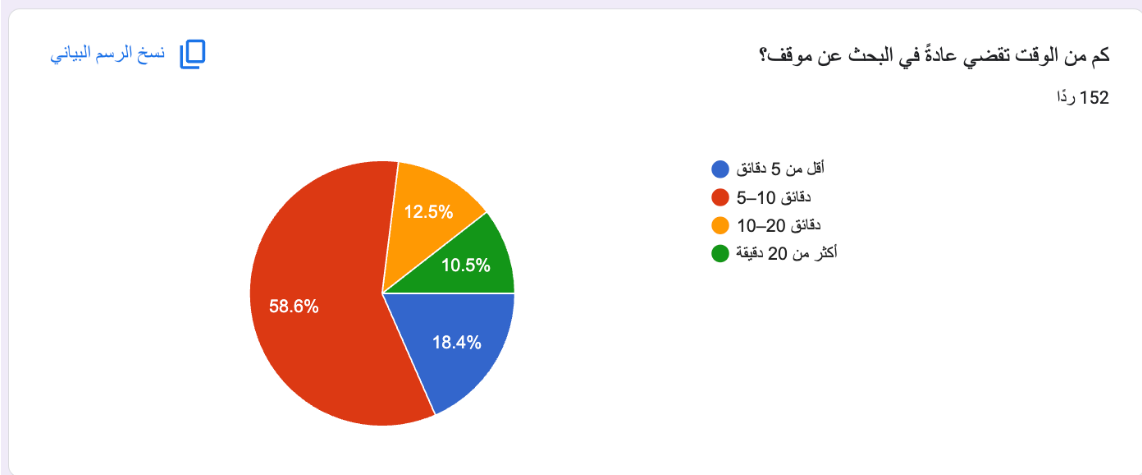


Figure 3. response survey



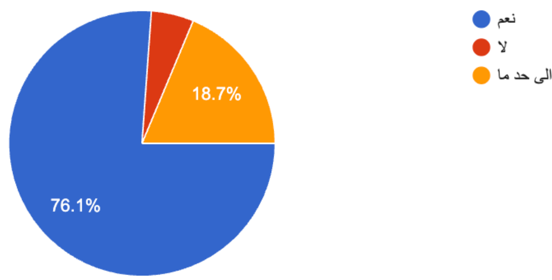

هل تواجه صعوبة في إيجاد موقف مناسب لسيارتك؟
153 ردًا
نسخ الرسم البياني
نعم
لا
احيانا
58.2%
13.7%
28.1%


كم من الوقت تقضي عادةً في البحث عن موقف؟
152 ردًا
نسخ الرسم البياني
أقل من 5 دقائق
5–10 دقائق
10–20 دقائق
أكثر من 20 دقيقة
58.6%
12.5%
10.5%
18.4%

هل تعتقد أن مواقف السيارات في مدينتك غير منظمة؟

155 ردًا

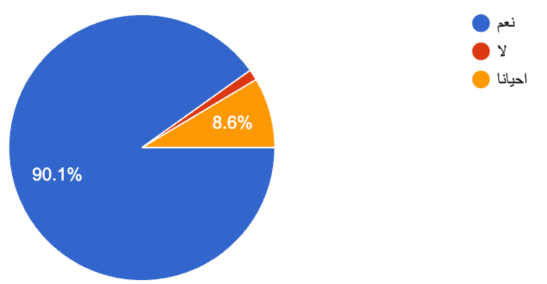
نسخ الرسم البياني



هل سبق أن واجهت مشكلة في عدم توفر مواقف في الأماكن العامة؟

152 ردًا

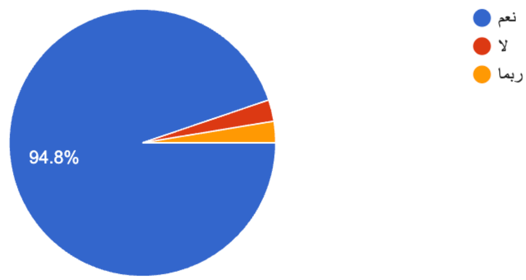
نسخ الرسم البياني



هل ترغب في وجود تطبيق يُظهر لك المواقف المتاحة في الوقت الفعلي؟

154 ردًا

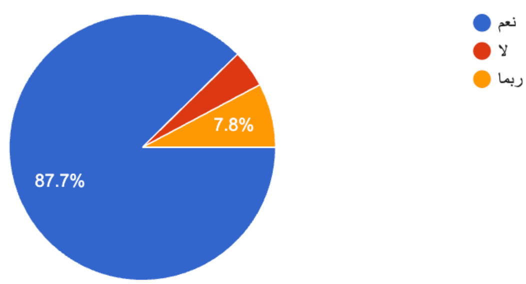
نسخ الرسم البياني



هل ترى أن إمكانية حجز موقف مسبقًا ستساعدك؟

154 ردًا

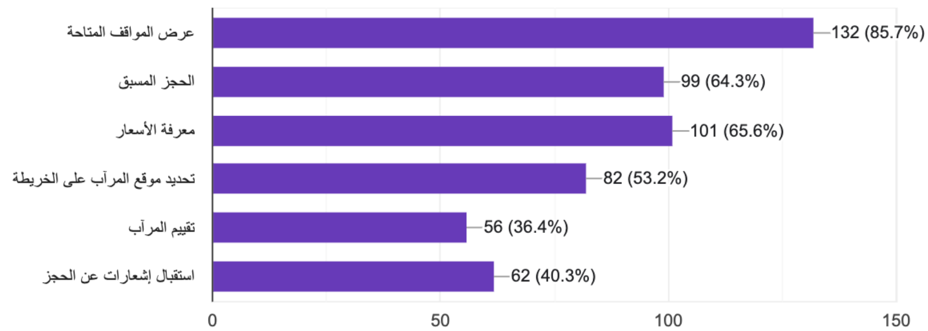
نسخ الرسم البياني



نسخ الرسم البياني

ما هي الميزات التي تهمك أكثر في تطبيق مواقف سيارات؟ (يمكن اختيار أكثر من خيار)

154 ردًا

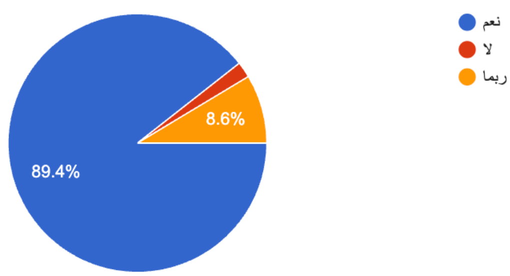


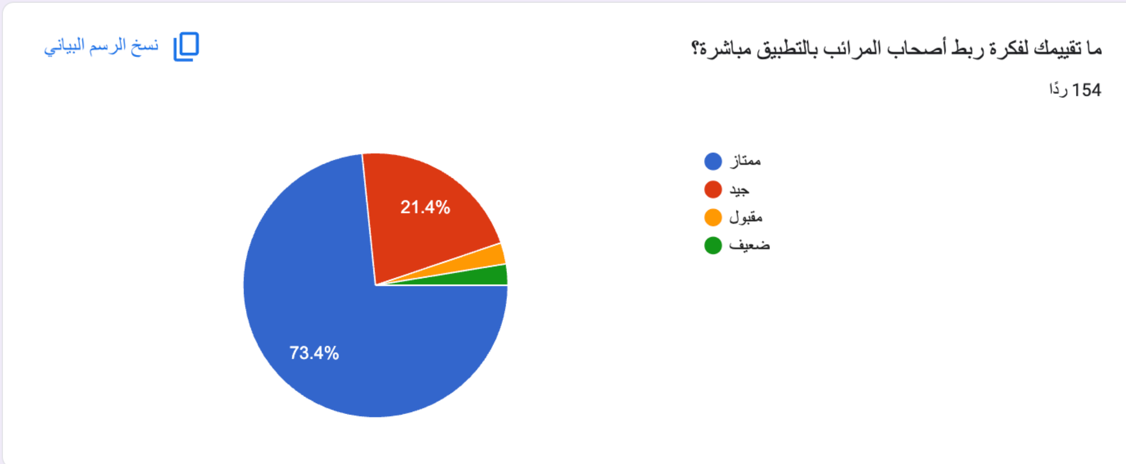
قسم بلا عنوان

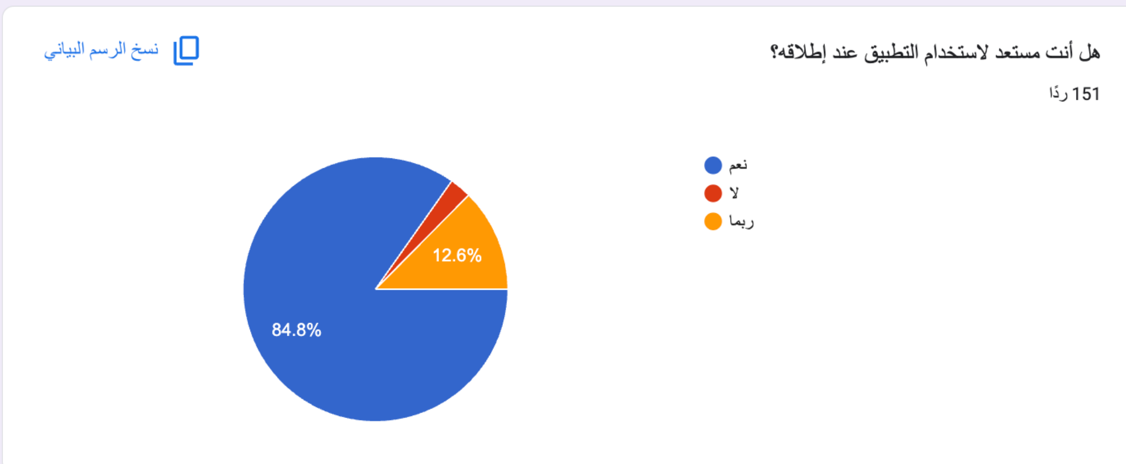
نسخ الرسم البياني

هل تعتقد أن مثل هذا التطبيق سيقلل الوقت الضائع في البحث عن موقف؟

151 ردًا




ما تقييمك لفكرة ربط أصحاب المرائب بالتطبيق مباشرة؟
154 ردًا
نسخ الرسم البياني
ممتاز
جيد
مقبول
ضعيف
21.4%
73.4%


هل أنت مستعد لاستخدام التطبيق عند إطلاقه؟
151 ردًا
نسخ الرسم البياني
نعم
لا
ربما
12.6%
84.8%

Feature ID	Stakeholder	Feature Name	Description	Priority (H/M/L)	Implementation Stage
F1	(Car Owner)	User Authentication	The system shall allow users to register and log in securely	H	Project 2
F2	(Car Owner)	View near Garages	The system shall allow users to view the garage based on the city	H	Project 2
F3	(Car Owner)	View parking availability	The system shall allow users to view the available place in the selected garage	H	Project 2
F4	(Car Owner)	View garage details.	The system shall allow users to view the garage information like (name, place, price, etc.	M	Project 2
F5	(Car Owner)	Reserve a Parking Spot	The system shall allow users to reserve the place in garage	H	Project 2
F6	(Car Owner)	Cancel Reservation	The system shall allow users to cancel the reservation 2 hours before the time.	H	Project 2
F7	(Car Owner)	View Reservation History	The system shall allow users to view their reservation history.	M	Project 2

Table 3. Functional requirement (Car Owner)

Feature ID	Stakeholder	Feature Name	Description	Priority (H/M/L)	Implementation Stage
F1	(Garage Owner)	User Authentication	The system shall allow users to log in securely	H	Project 2
F2	(Garage Owner)	Add the garage information	The system shall allow the user to add the garage details like name, location, price,etc.	H	Project 2
F3	(Garage Owner)	Update garage availability	The system shall allow users to add available place in garage	H	Project 2
F4	(Garage Owner)	Manage Reservations	The system shall allow garage owners must be able to: View reservations, Accept or deny reservation requests	H	Project 2
F5	(Garage Owner)	View Reservation History	The system shall allow garage owners to view their reservation history.	M	Project 2
F6	(Garage Owner)	Edit Garage Information	The system shall allow garage owners to be able to update pricing, capacity, and working hours.	H	Project 2

Table 4Functional requirement (Garage Owner)

Feature ID	Stakeholder	Feature Name	Description	Priority (H/M/L)	Implementation Stage
F1	(Admin)	User Authentication	The system shall allow admin to log in securely	H	Project 2
F2	(Admin)	Add the garage owner	The system shall allow the admin to add the garage owner that will be on our application.	H	Project 2
F3	(Admin)	Manage Users	The system shall allow Admin to view, approve, deactivate, or delete user accounts.	H	Project 2
F4	(Admin)	Manage Garages	The system shall allow Admin to approve garages before they appear to users.	H	Project 2
F5	(Admin)	View System Reports	The system shall allow admin to view a report about the system overall.	M	Project 2

Table 5. Functional requirement (Admin)

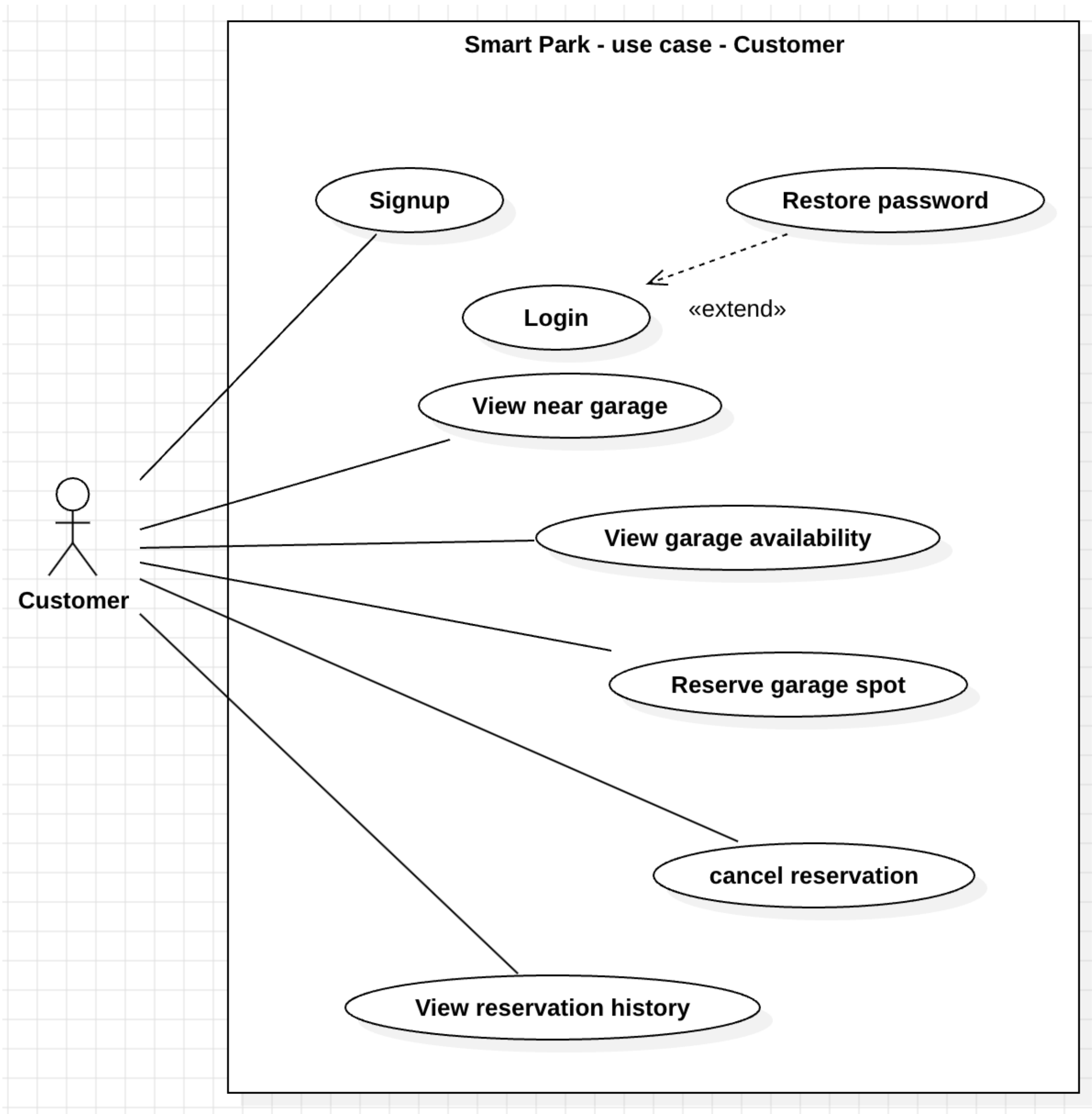


Figure 4. use case - Customer

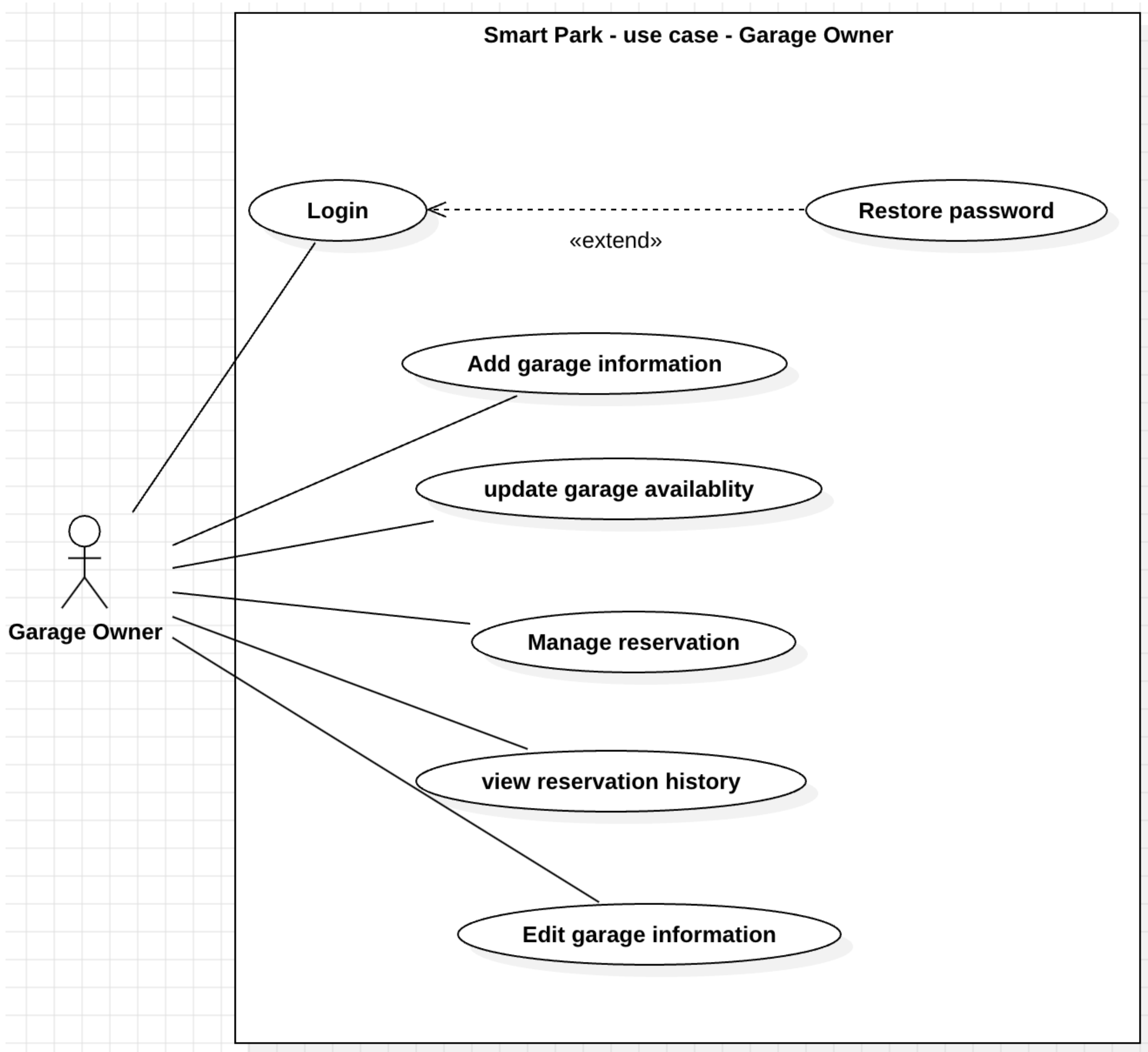


Figure 5. use case - garage owner

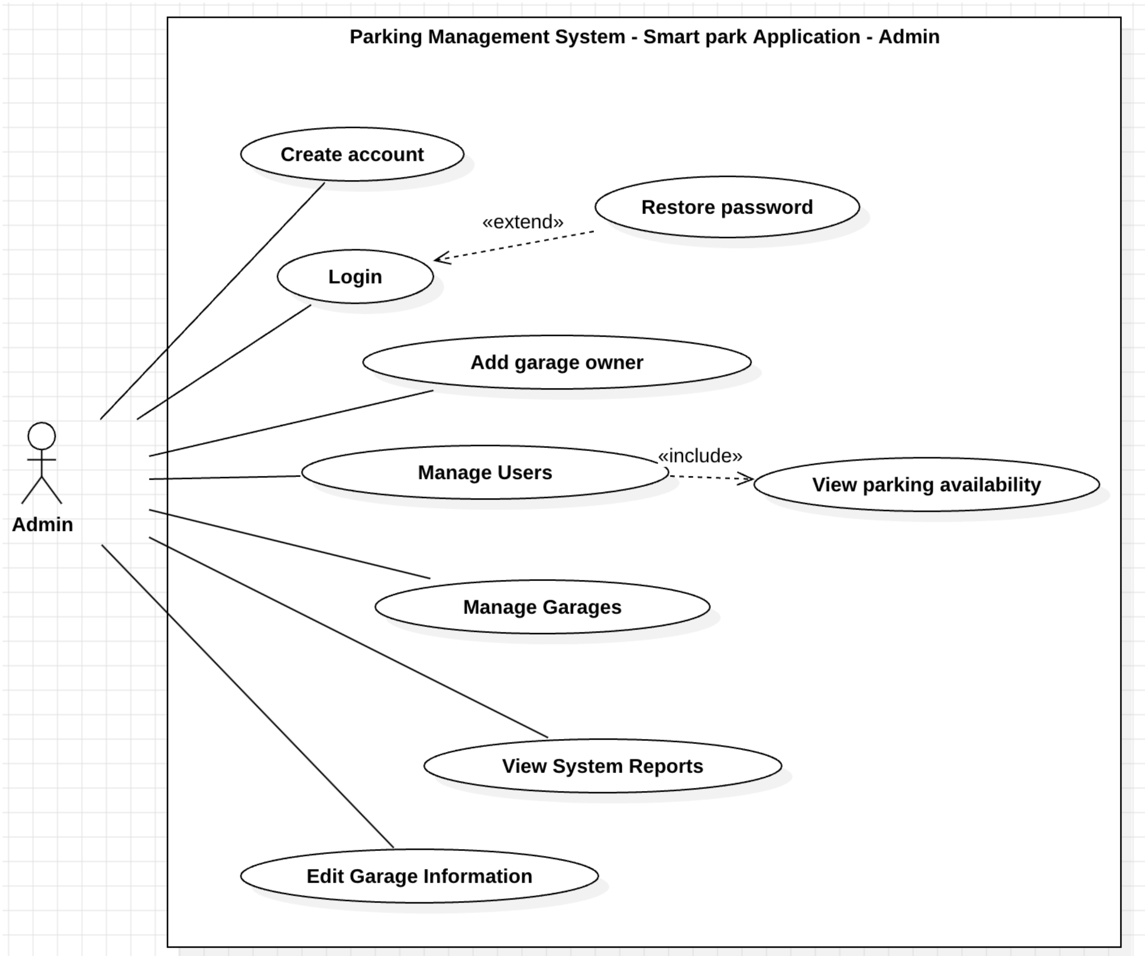


Figure 6. use case - admin

2.2 Feature-to-Requirement Mapping

Non-Functional Requirement ID	Requirement Description	Priority
NFR-1 (Security)	-All passwords must be stored in the database using a secure hashing algorithm. -The login process shall be completed within 3 seconds under normal network conditions.	H

Table 6. non-Functional requirement

Non-Functional Requirement ID	Requirement Description	Priority
NFR-2 (Availability)	-The system shall be available 24/7 times. -The system shall be available on android and IOS.	H

Table 7.non-Functional requirement

Non-Functional Requirement ID	Requirement Description	Priority
NFR-3 (Usability)	-The system shall be useable. -The system shall be provided Arabic and English language	H

Table 8.non-Functional requirement

Non-Functional Requirement ID	Requirement Description	Priority
NFR-4 (performance)	-The system shall respond for any request within (2-3) seconds -The system shall provide updates on user's information in maximum of (2-3) second	H

Table 9. non-Functional requirement

Non-Functional Requirement ID	Requirement Description	Priority
NFR-5 (maintainability)	-The system shall be maintained by the developers.	H

Table 10. non-Functional requirement

Non-Functional Requirement ID	Requirement Description	Priority
NFR-6 (Reliability)	-The system shall keep backup data for user information. -The system shall work 100% of the time.	M

Table 11. non-Functional requirement

CHAPTER 3: System Design and Deployment Overview

3.1 System Architecture

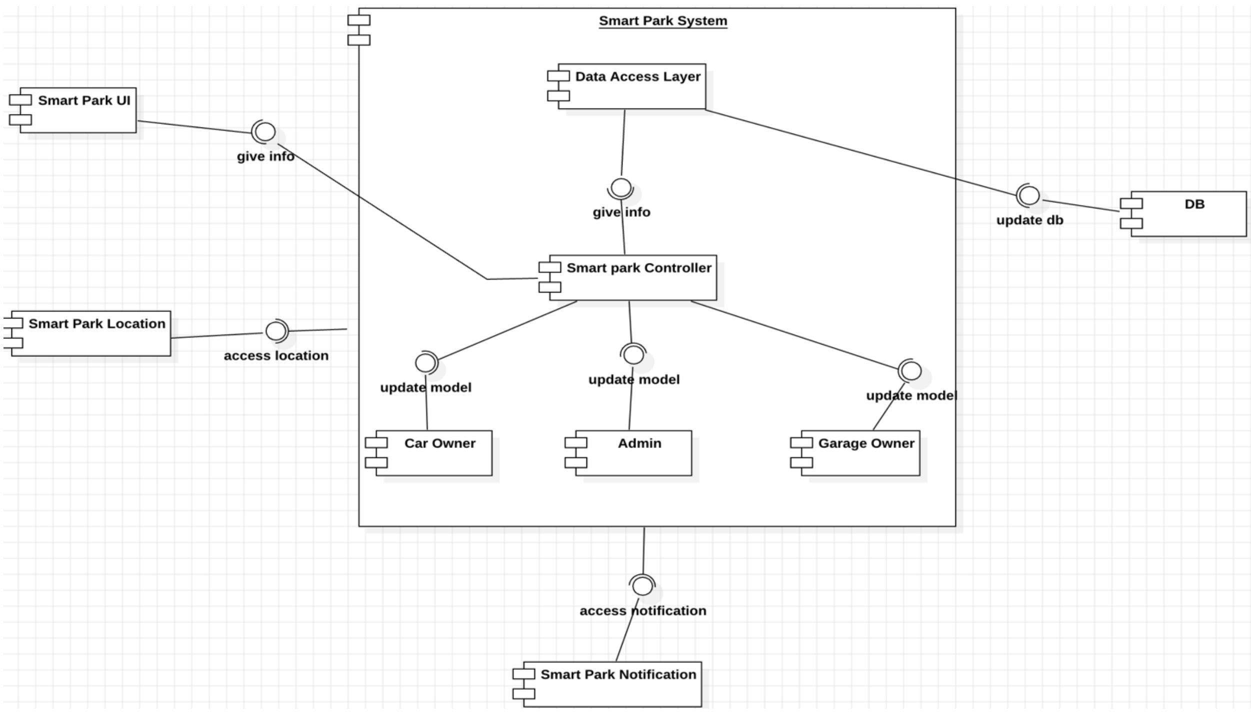


Figure 7. component diagram

The Smart Park system follows a layered, component-based architecture. The presentation layer consists of the mobile components (Smart Park UI, Location, and Notification), which interact only with the Smart Park Controller. The controller acts as the API and coordination layer, receiving requests from the mobile app and delegating them to the backend business modules: Car Owner, Garage Owner, and Admin. These modules contain the logic for managing users, garages, reservations, and administration tasks. All data access is centralized through the controller, which communicates with the database component (DB). This separation of concerns improves maintainability, scalability, and flexibility of the system.

3.2 Detailed Design (UML Models Based on Implementation)

Class Diagram:

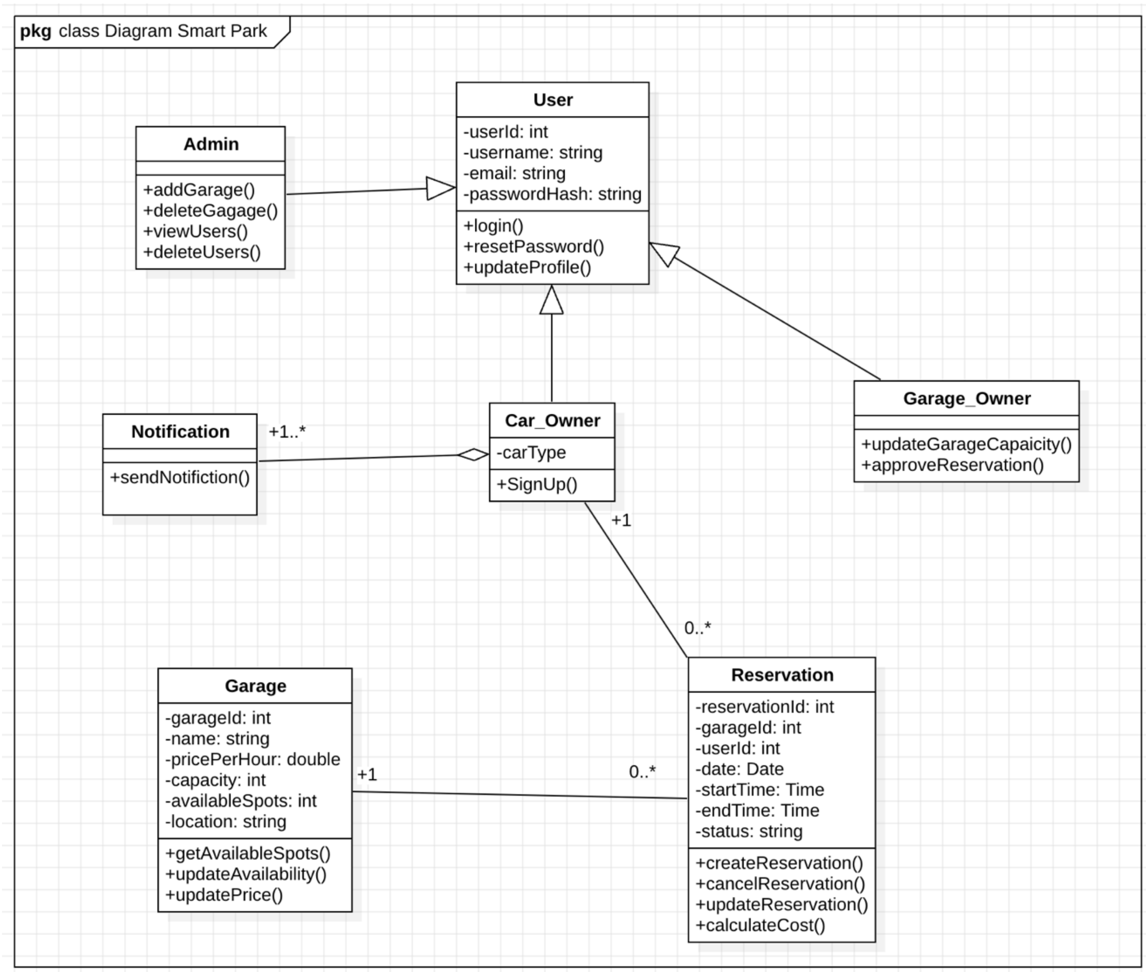


Figure 8. Class Diagram

Sequence Diagrams:

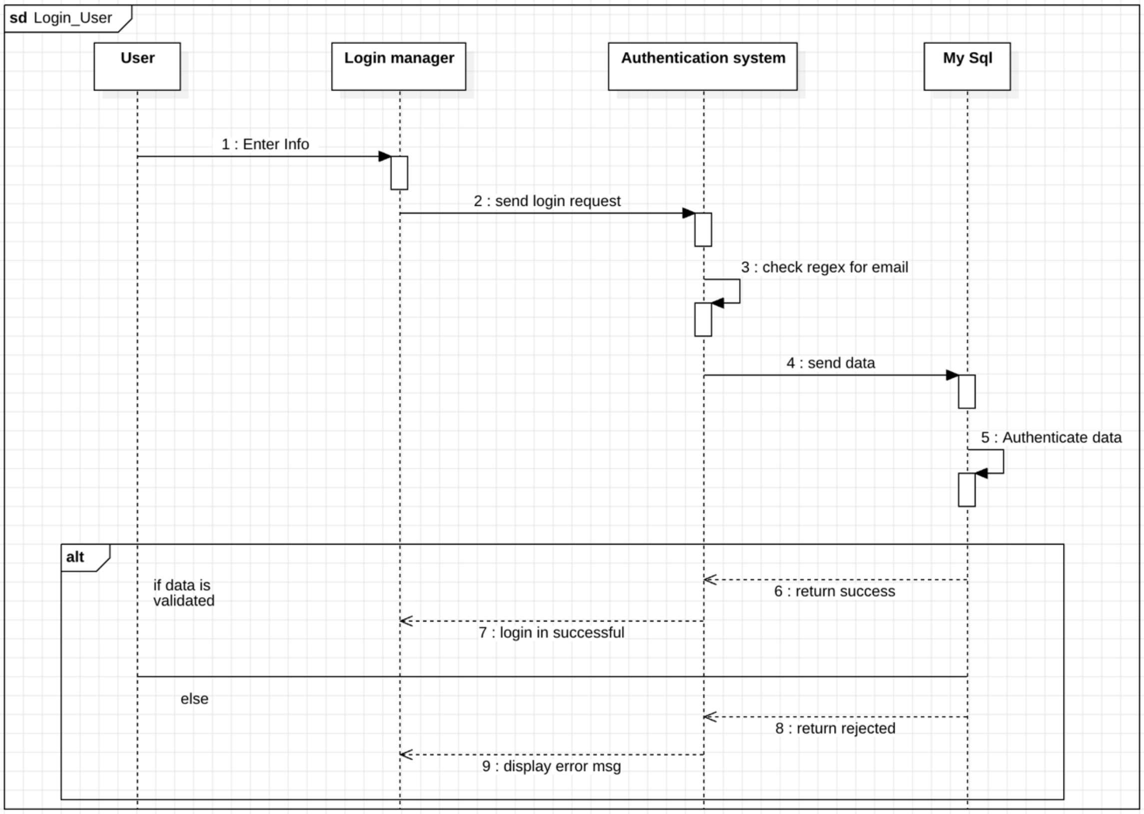


Figure 9. sequence Diagram

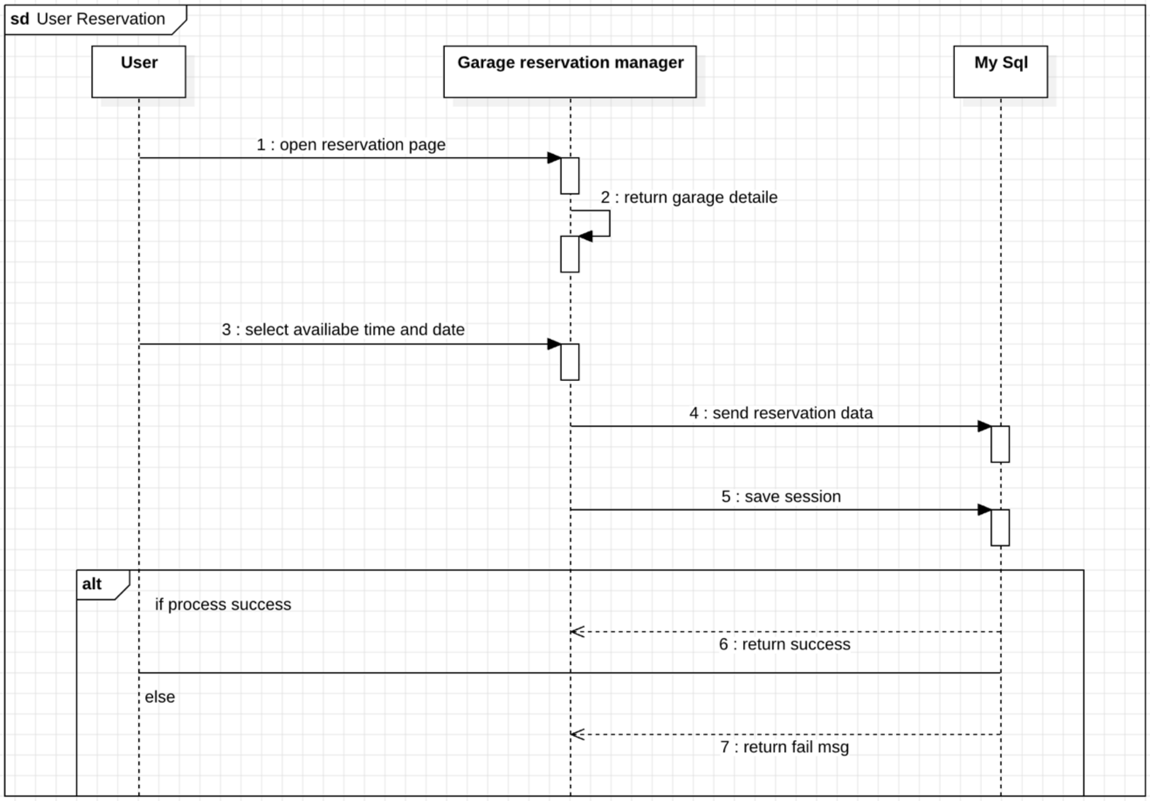


Figure 10. Sequence Diagram

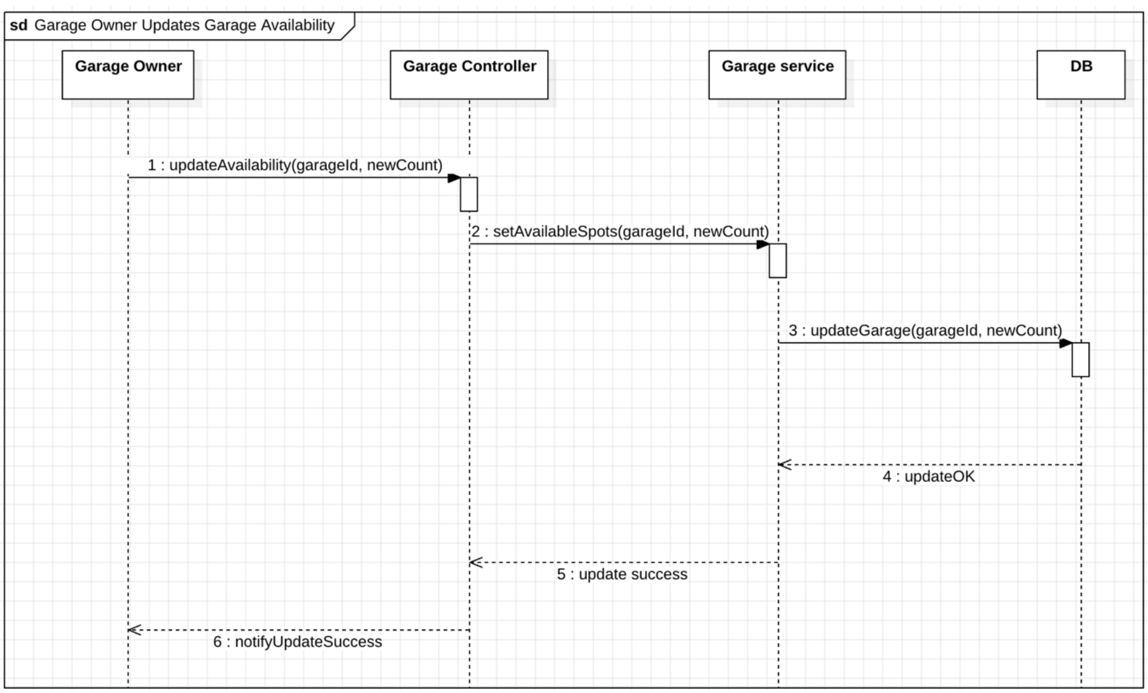


Figure 11. Sequence Diagram.

State Machine Diagram:

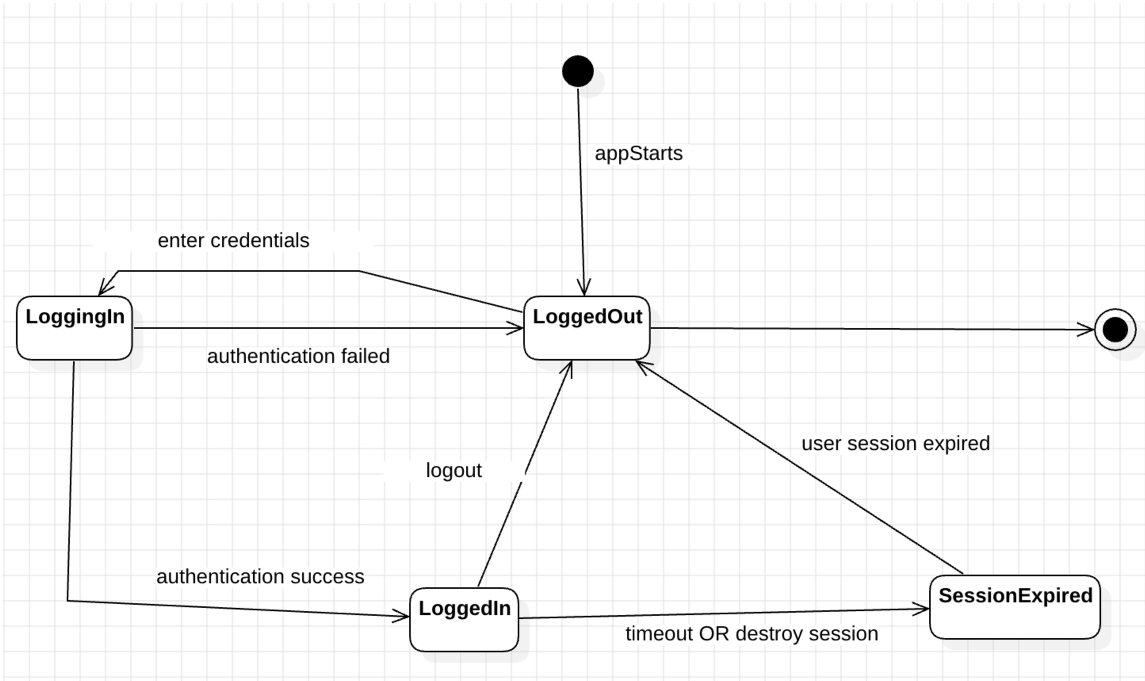


Figure 12. State Machine login.

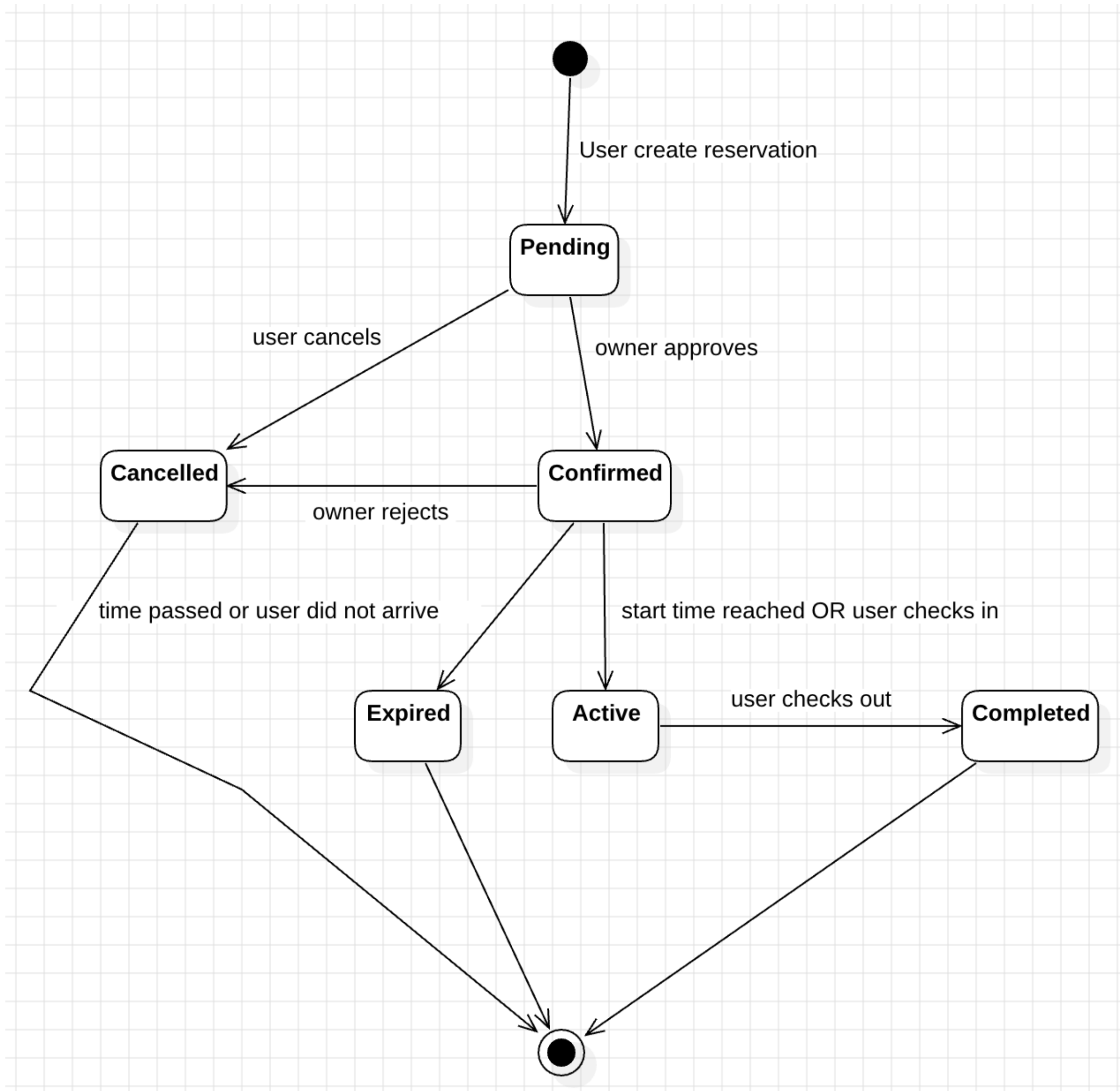


Figure 13. State Machine Reservation.

3.3 Software Deployment

Component	Technology	Purpose
Front-End	Flutter framework	Used to build the mobile application for Android and iOS with a single codebase. Provides fast UI development, smooth animations, cross-platform support, and native performance.
Back-End	PHP Laravel Framework	Handles all server-side logic such as authentication, reservations, garage management, and API creation. Provides routing, middleware, security, and MVC architecture for clean backend development.
Database	MySQL	Stores all application data including users, garages, reservations, availability, and history. Reliable, scalable, and widely supported relational database.
Dev Tools	Github, Android studio, VS Postman	Main IDE for testing and running the Flutter mobile app on Android emulators and physical devices. Helps debug and profile mobile app performance.

Table 12. Software Deployment

CHAPTER 4: System Development and Implementation

4.1 Core Implementation Progress

This chapter presents the development progress of the Smart Park system during GP1. The objective of this phase was to implement the core functionality of the application and establish a stable foundation for the remaining components that will be completed in GP2. The development focused on implementing the essential mobile application features using Flutter and building the backend services using Laravel, MySQL, and RESTful API communication.

Several key features of the system have been successfully implemented. The user authentication module, including registration and login, is fully functional and communicates securely with the backend API. Users can browse available garages within a selected city, view detailed information about each garage—such as name, location, price per hour, capacity, and available parking spots—and initiate a reservation request. The initial reservation workflow is operational and stores reservation information in the backend database through Laravel controllers and models.

The backend architecture has been structured according to MVC principles, and multiple API endpoints have been developed to manage users, garages, and reservations. The database schema has been implemented using MySQL, and phpMyAdmin was used for database management and testing. The system currently supports core CRUD operations, authentication handling, and data retrieval for the mobile application.

Some components are partially implemented, including real-time parking availability updates, reservation history, garage capacity updates, and cancellation constraints. Other components—such as the admin web dashboard, garage owner dashboard, and additional management tools—are planned for development during GP2. These upcoming modules will build on the foundations established in GP1.

GitHub Repository: Link

4.2 Implemented and Planned Features

Feature ID	Feature Name	Related Requirement	Implementation Status	GitHub Link
F1	(User Authentication) Login and Signup	FR-01, FR-02	Implemented	[Link]
F2	View Nearby Garages	FR-03	Implemented	[Link]
F3	View Garage Details	FR-04	Implemented	[Link]
F4	Parking Availability	FR-05	To-be-implemented-GP2	-
F5	Reserve Parking Spot	FR-06	Implemented	[Link]
F6	Cancel reservation	FR-07	To-be-implemented-GP2	-
F7	Add Garage Information (Garage Owner)	FR-08	To-be-implemented-GP2	-
F8	Manage Reservations (Garage Owner)	FR-11	To-be-implemented-GP2	-
F9	Admin Manage Users	FR-15	To-be-implemented-GP2	-
F10	Admin Manage Garages	FR-16	To-be-implemented-GP2	-
F11	Notifications	FR-20	To-be-implemented-GP2	-

Table 13. Implemented and Planned Features

Test cases:

Test cases F1	TC	Valid-invalid
Login	Email: Test@gmail.com Password: Test123@.	valid
Login	Email: Test2002@gmail.com Password: Test123@.	Invalid – this email not registered.
Signup	Name: Mohamed Email: mohamed@gmail.com Password: Mohamed123@. Confirm-password: Mohamed123@.	Valid – registered done successfully.
Signup	Name: Mohamed Email: mohamed@gmail.com Password: Mohamed123@@ Confirm-password: Mohamed123@.	Invalid – the password and confirm password does not match

Table 14. Test cases

Test cases F5	TC	Valid-invalid
Reserve Parking Spot	Number of spots: 2 Start date: 20/1/2026 End date: 20/1/2026 Start time: 10:00 AM End time: 12:00 AM	Valid – spots reserved
Reserve Parking Spot	Number of spots: 2 Start date: 20/1/2026 End date: 20/1/2026 Start time: 10:00 AM End time: 08:00 AM	invalid – end time must after start time.

Table 15. Test cases

4.3 Screenshots Evidence

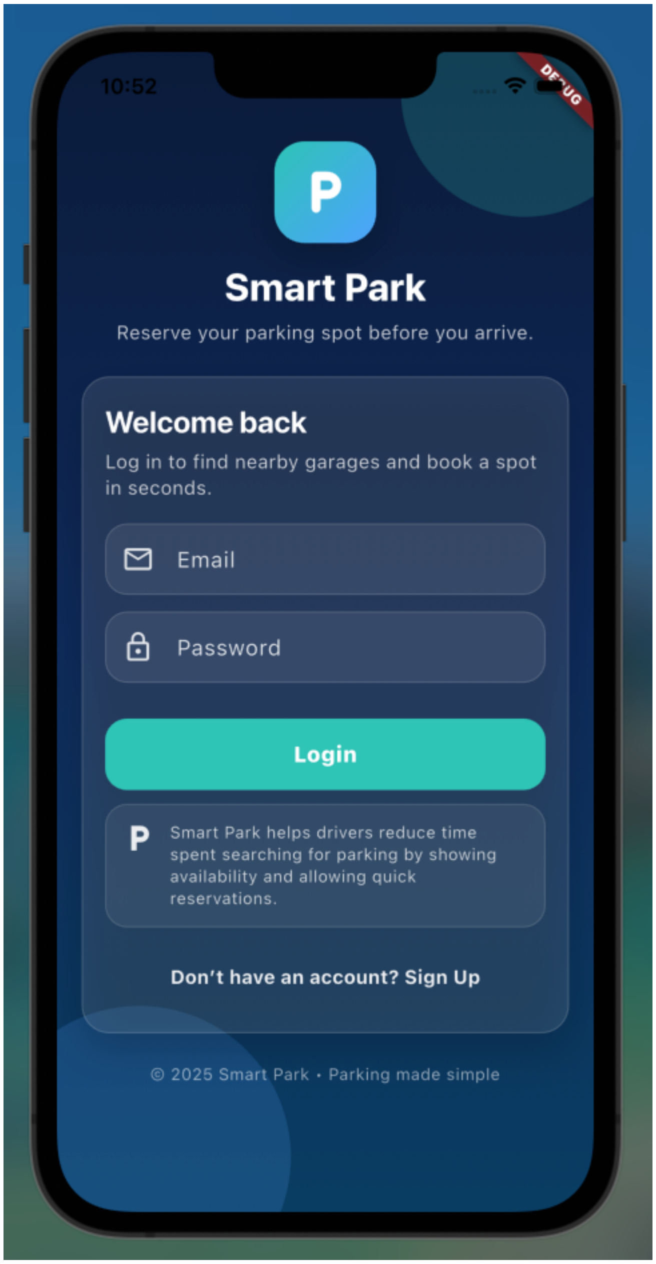


Figure 15. Login screen

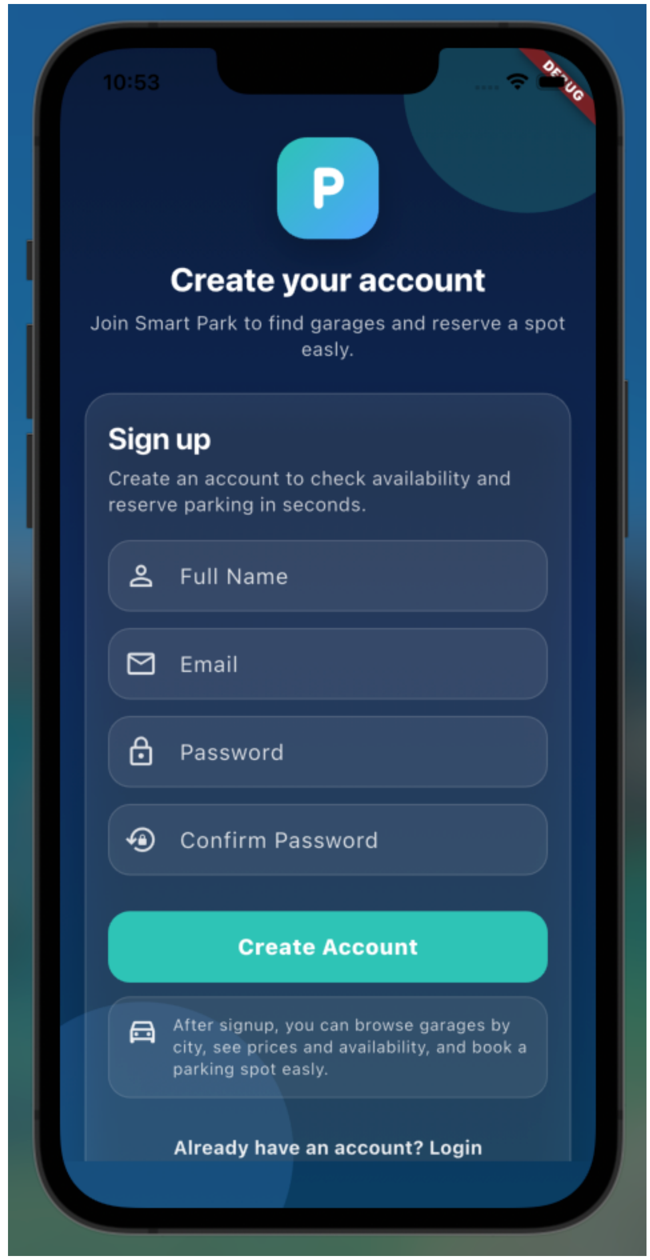


Figure 14. Signup screen.

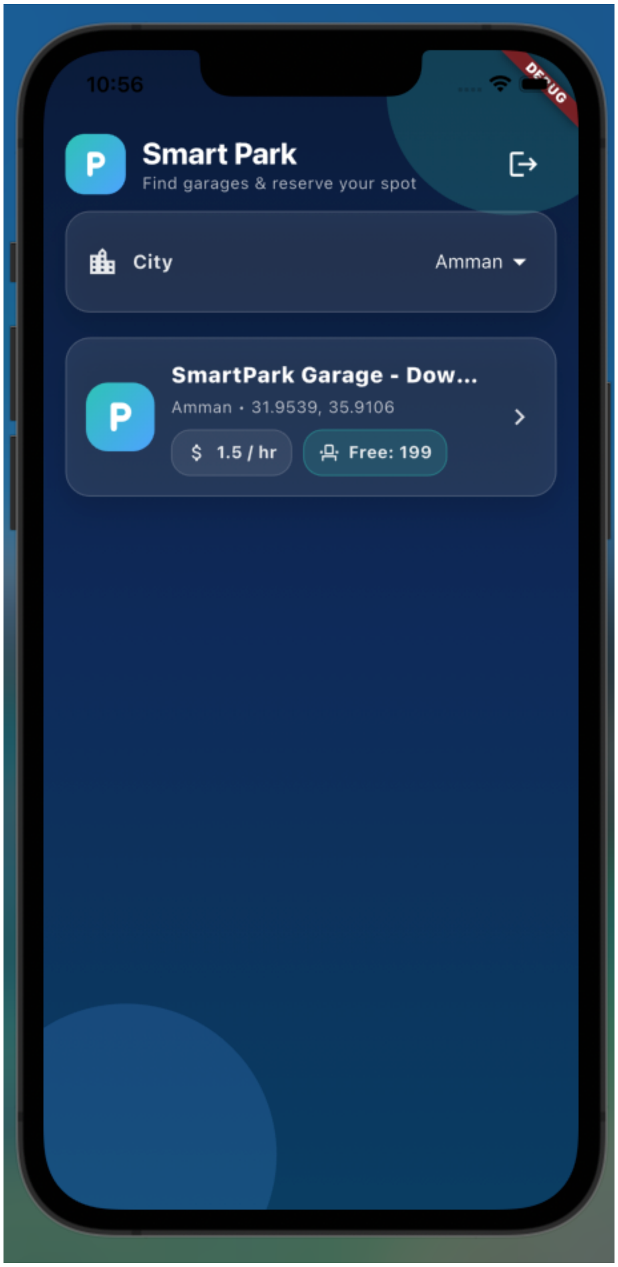


Figure 17. Home screen

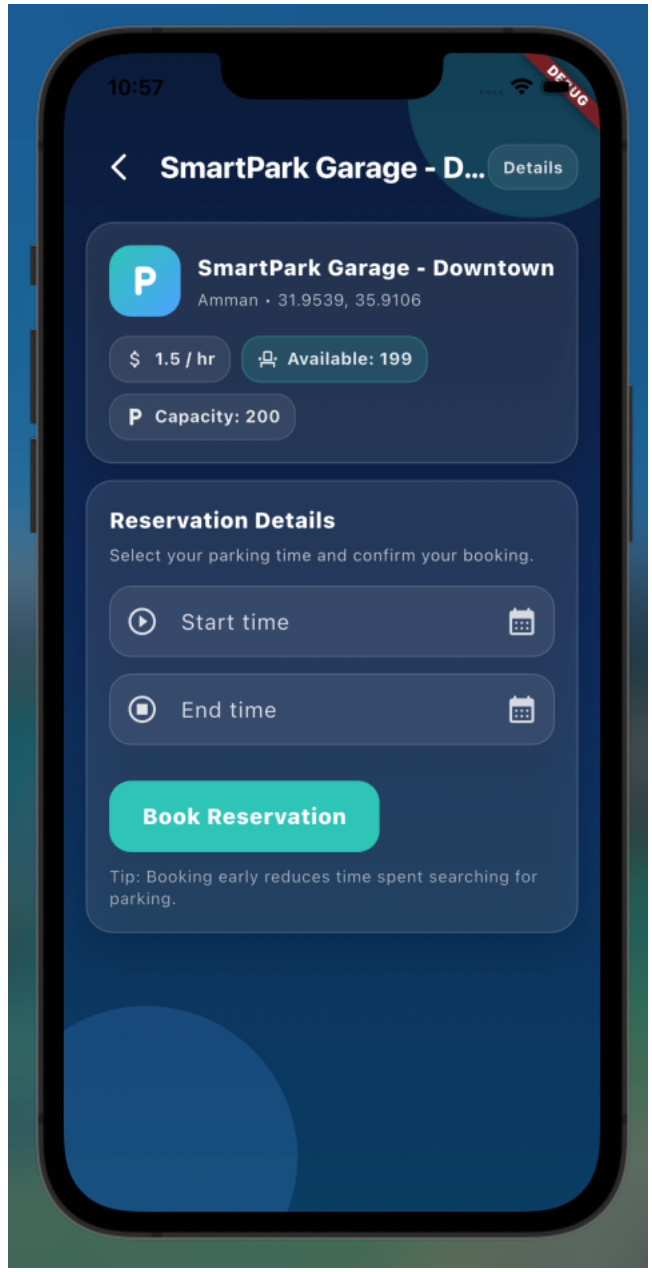


Figure 16. garage details.

CHAPTER 5: Testing

5.1 Testing Overview

The purpose of testing in GP1 was to verify that the implemented components of the Smart Park system function correctly, meet their defined requirements, and interact reliably between the mobile application and the backend API. Testing ensured that the core features—such as user authentication, viewing nearby garages, retrieving garage details, and submitting reservation requests—operate as expected and support the foundation for further development in GP2.

Several types of testing were performed during this phase. **Unit testing** was conducted on backend Laravel controllers and models to validate database operations and input validation. **Integration testing** was performed to ensure correct communication between the Flutter frontend and the Laravel API. **API testing** using Postman was used to verify request/response correctness, status codes, and error handling. Finally, **UI testing** was performed manually to evaluate the mobile application screens, interactions, and user flow.

Testing Scope:

The following modules were covered during GP1 testing:

- User Registration API
- Login Authentication API
- Mobile UI flow for login and signup.
- Database validation for insertions and updates

Tool / Framework	Purpose
Postman	API testing for Laravel endpoints (login, view garages, reservations)
Laravel PHPUnit	Unit and integration tests for backend controllers and models
Flutter DevTools	UI debugging and widget validation
MySQL / phpMyAdmin	Verifying database entries during testing
Android Studio Emulator	Running and testing the mobile application

Table 16. testing scope

5.2 Sample Test Case

Test Case ID	Feature	Input	Expected Output	Actual Output	Result	Evidence / Link
TC-01	Login	Valid credentials	User is authenticated, redirected to home	Home screen loaded	Pass	App screenshot (found it below)
TC-02	Login	invalid credentials	Error message: “Invalid credentials”	Error shown	Pass	App screenshot (found it below)
TC-03	Signup	New user data	Success message: “Signup done successfully” and move to login page	Login page loaded	Pass	App screenshot (found it below)
TC-04	Signup	New invalid user data	Error message: “User data is not correct with exactly error”	Error shown	pass	App screenshot (found it below)
TC-05	View Garage Details	Garage ID	Return garage information	Details view correctly	Pass	Postman screenshot
TC-06	Create Reservation	User id, garage id, time	Reservation store in data base	Reservation saved	Pass	DB screenshot
TC-07	Create Reservation	Missing parameter	Error: “invalid input”	Error shown	Pass	API response

Table 17. sample test case

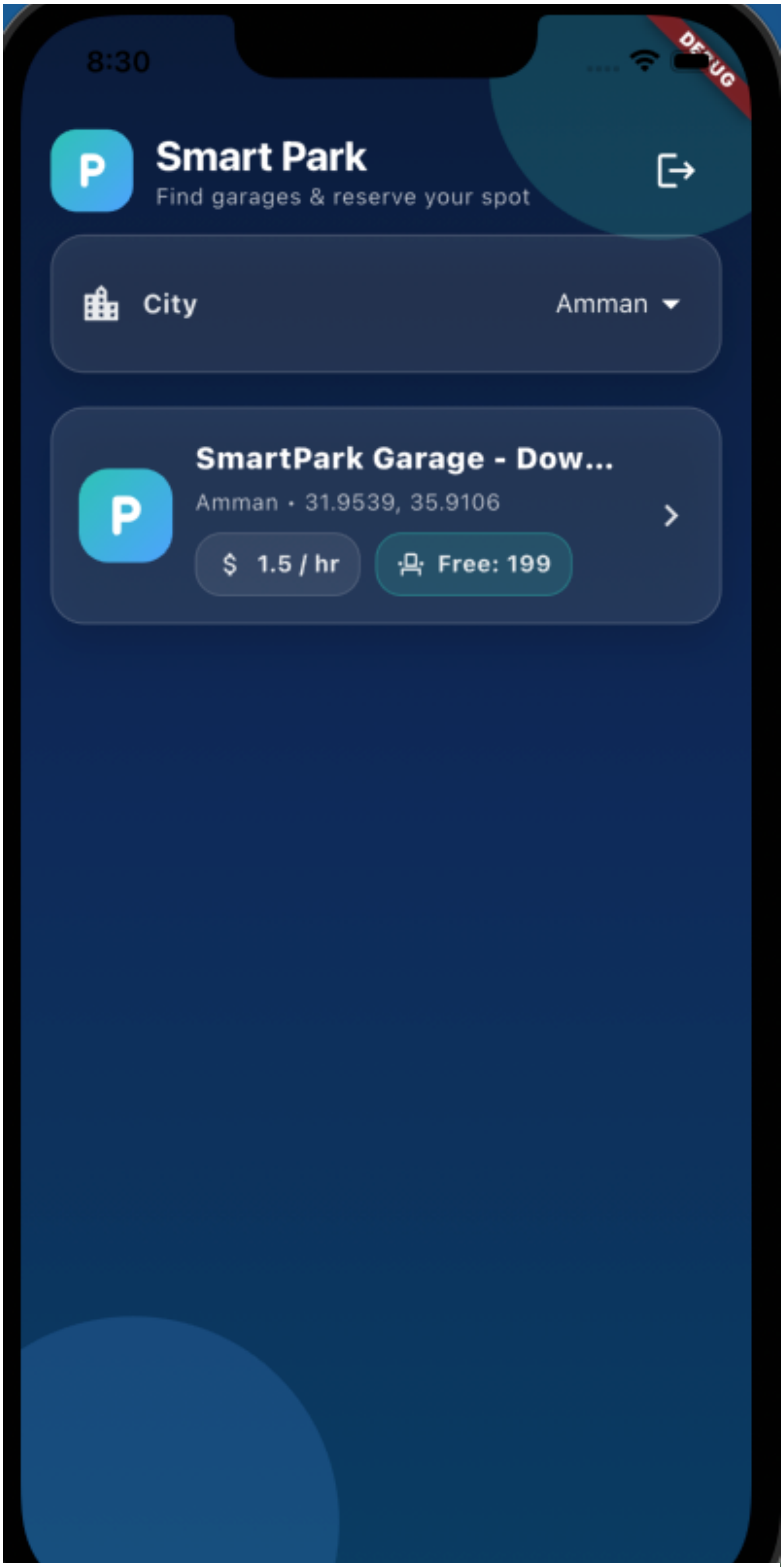


Figure 18. vaild login attempt

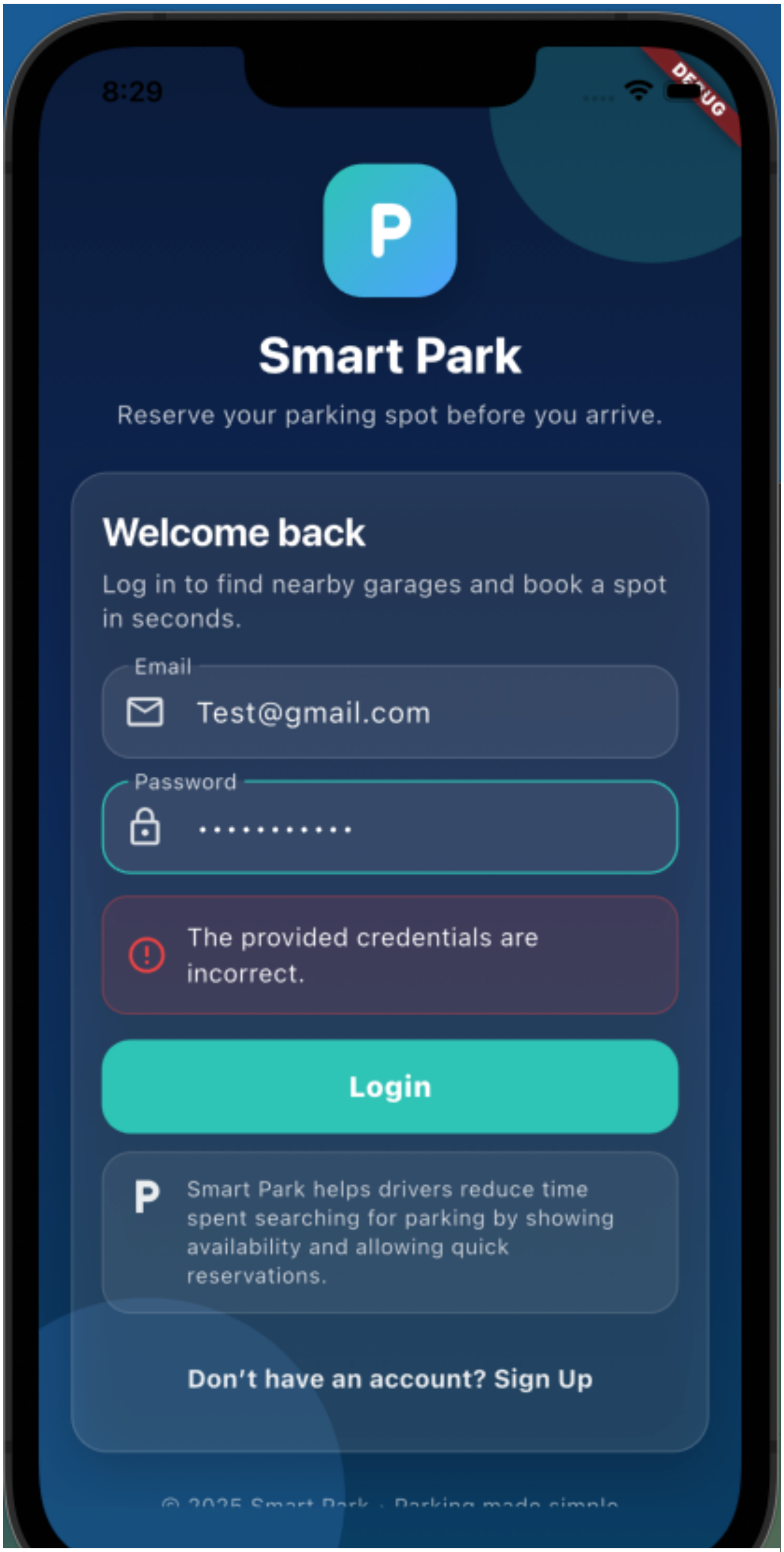


Figure 19. Invalid Login attempt

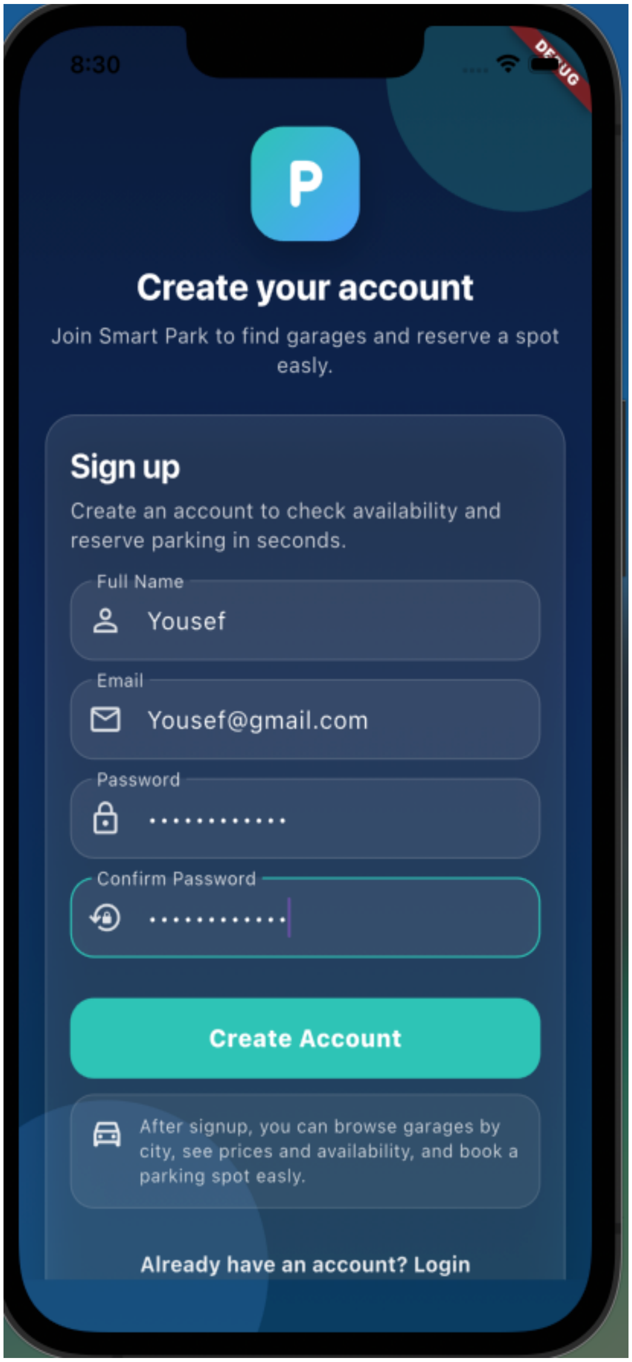


Figure 20. insert data for valid signup.

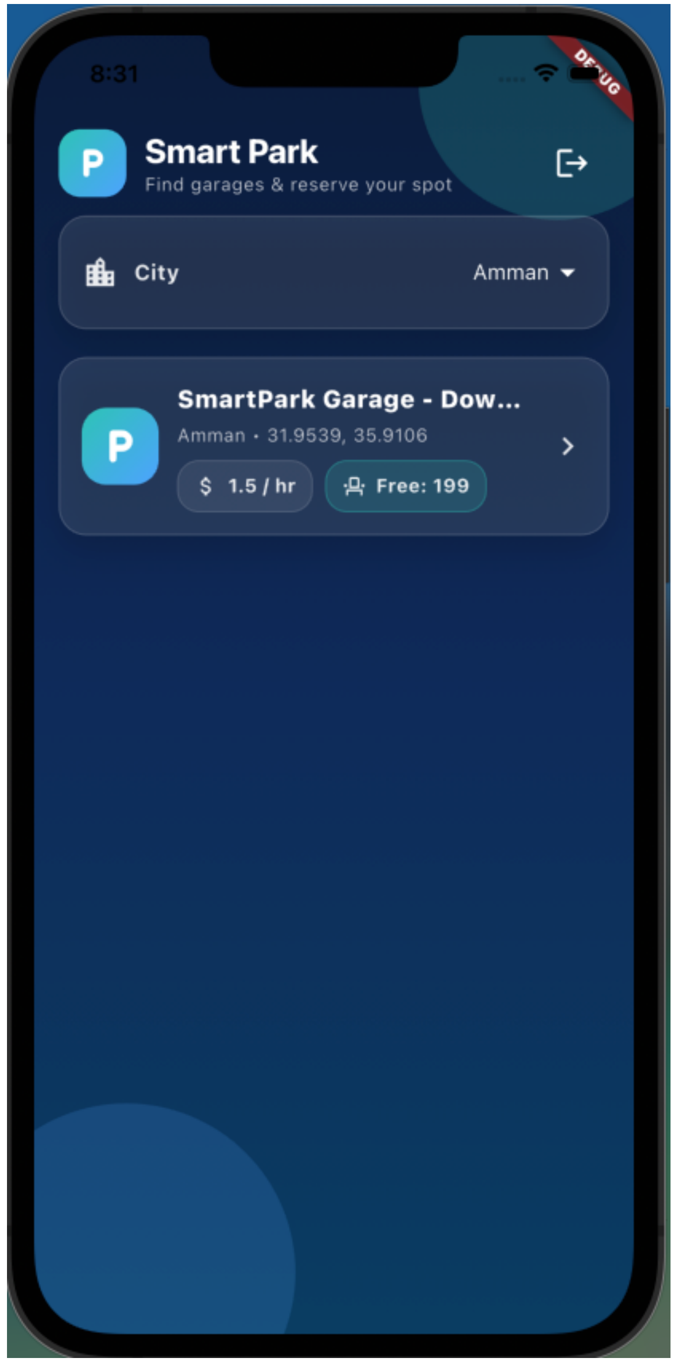


Figure 21. after signup go to home page

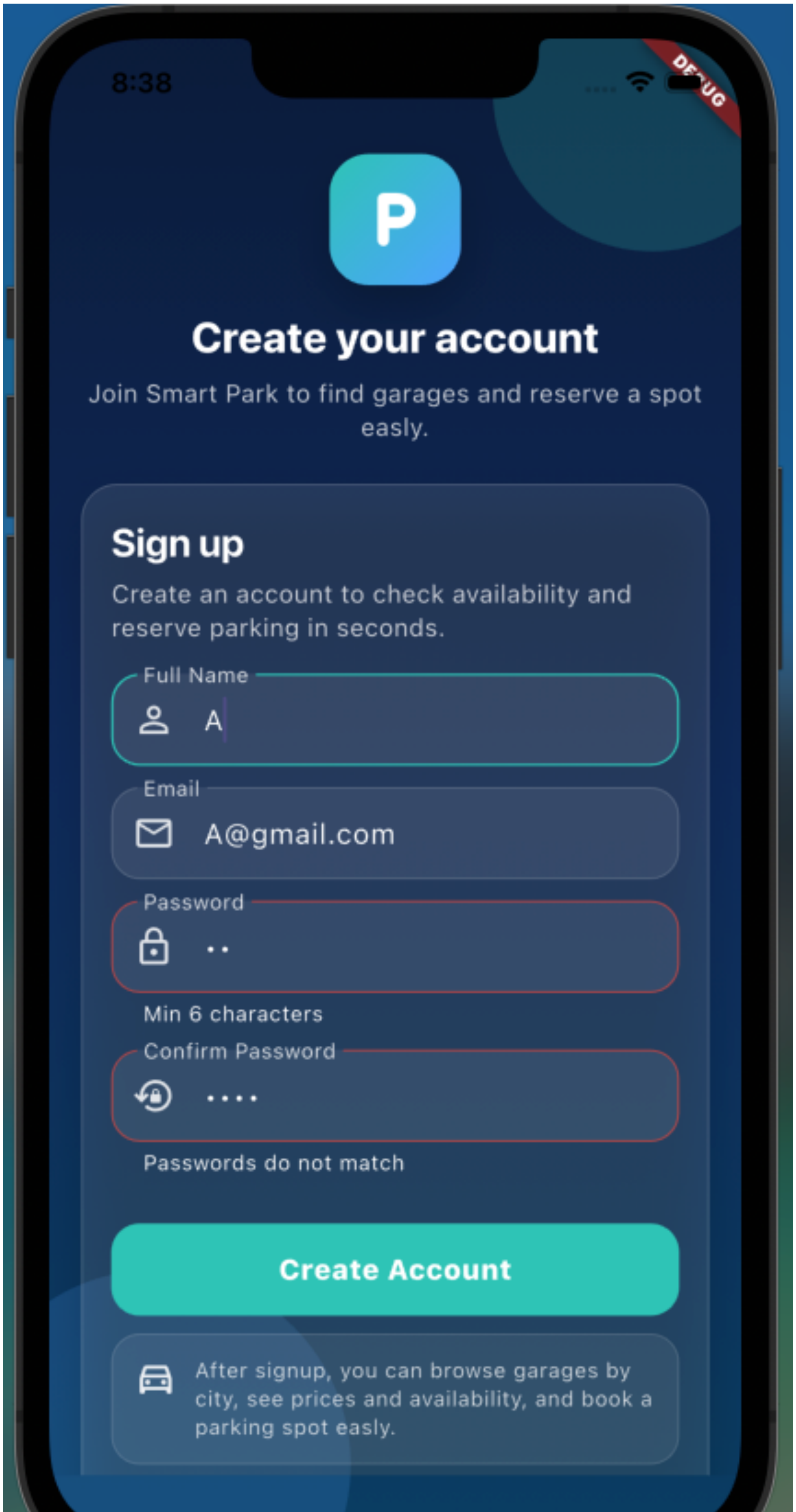


Figure 22. insert data for invailed sginup attempt

Figure 23. View Garage Details

Figure 25. Create Reservation

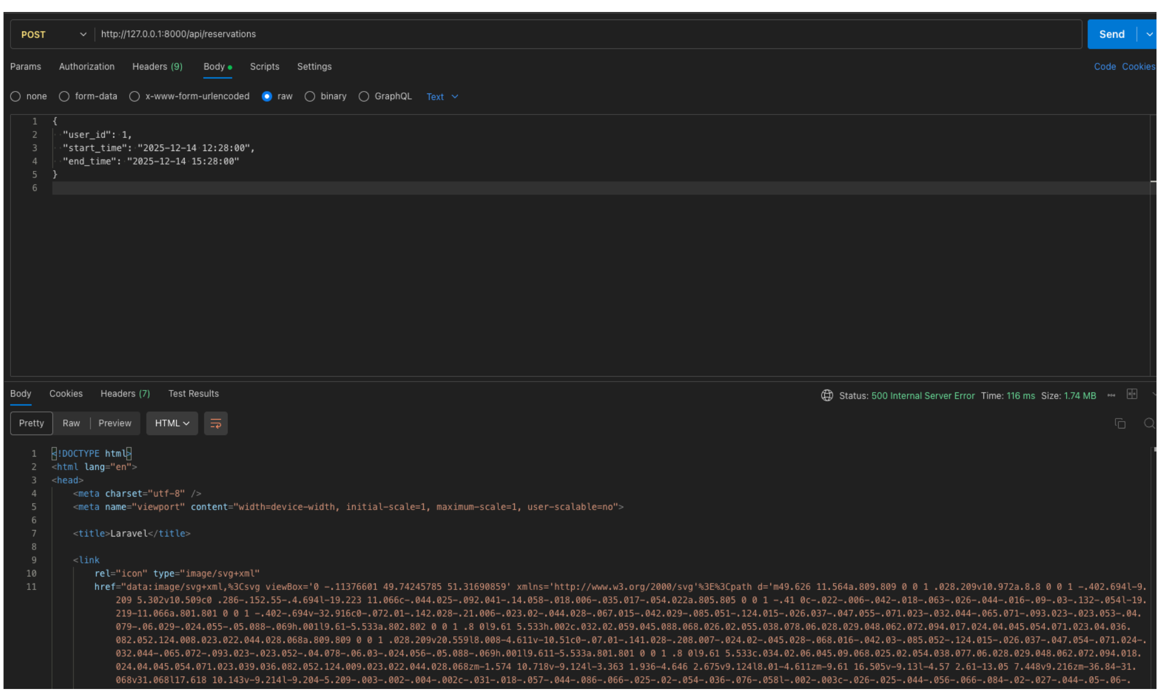


Figure 24. TC-7 missing garage_id

5.3 Test reports

API Test Results

Using Postman, each endpoint (login, register, get garage details,) was tested for:

- Success responses (200 status)
- Error responses (404)

Figure 26. vailed login screen.

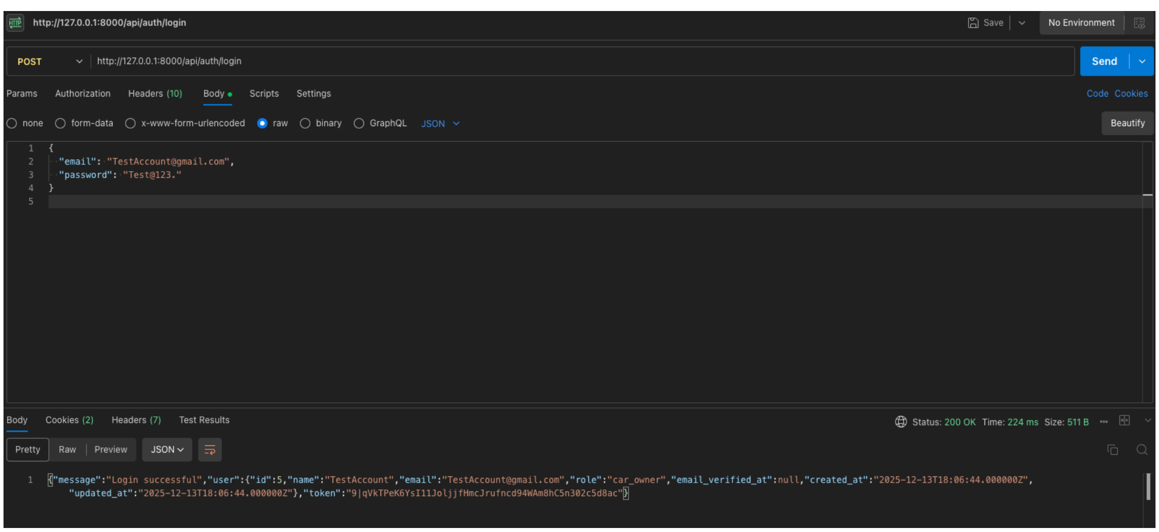


Figure 27. valied signUp screen.

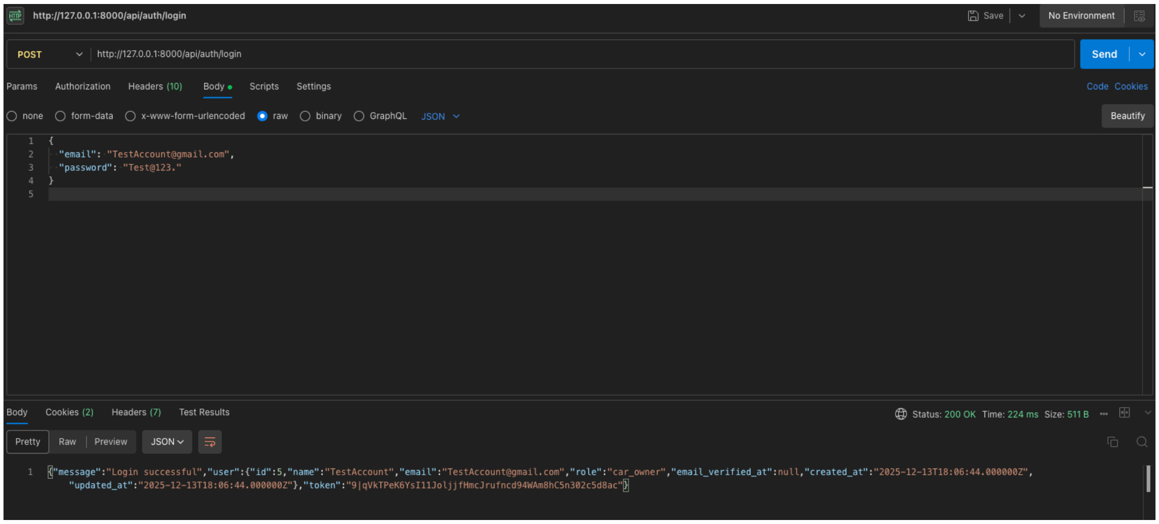


Figure 28. get garage details

Backend Unit Test Results

Laravel PHPUnit tests validated:

- User authentication logic
- Get garage detailes.
- Reservation creation logic

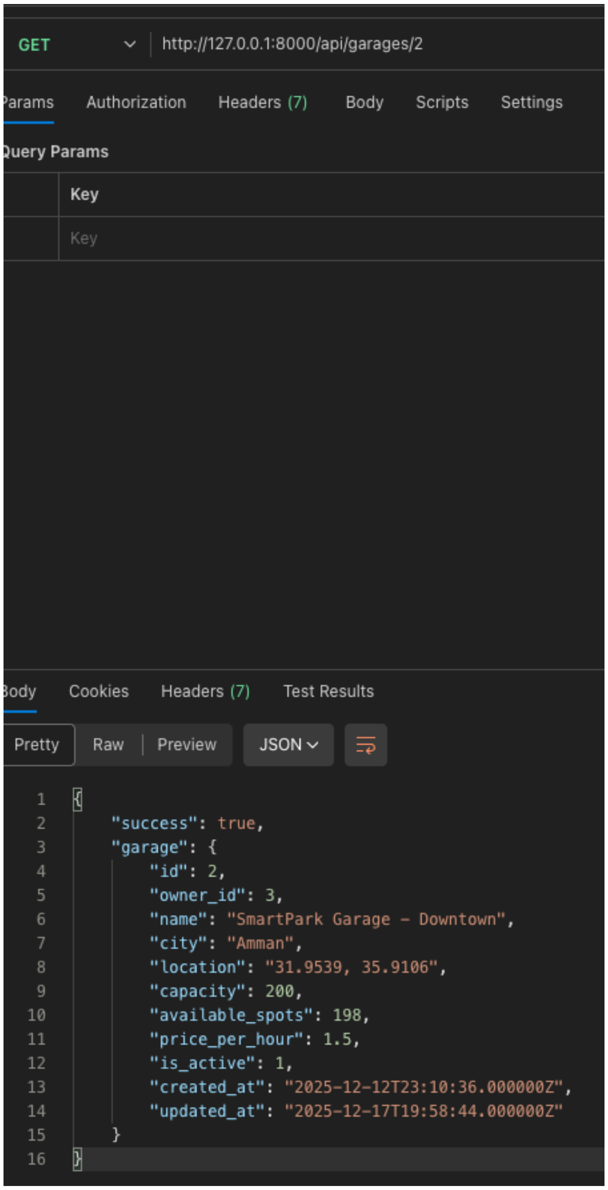


Figure 30. get garage details

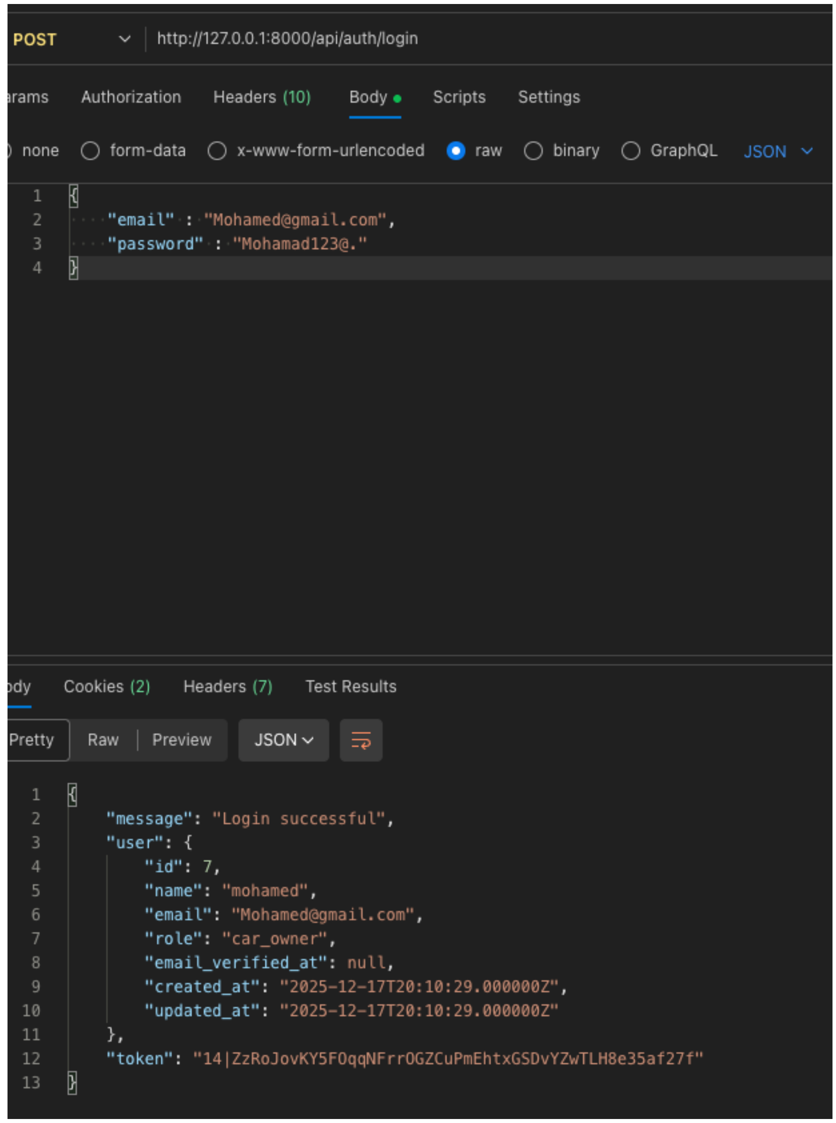


Figure 29. login postman

UI and Integration Test Results

Using the androied emulator:

- The login flow was tested successfully
- The garage list screen loads the correct data
- The garage details screen shows all required fields
- Sending a reservation request reaches the backend and stores correctly.

References

List all frameworks, APIs, libraries, and research papers used.